IDSMS-CIS: Internship Document Submission and Monitoring System for CIS

An IT Proposal Presented to the
Faculty of the Information Technology Department
School of Accountancy, Management, Computing and Information Studies
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IT 321

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Abstract

Internship programs in higher education will require structured documentation, monitoring, and compliance tracking to ensure that students meet academic and industry requirements. At Saint Louis University, Google Classroom will serve as the primary platform for internship document submission; however, it lacks integrated compliance monitoring features, leading to fragmented tracking, limited visibility, and increased administrative workload.

The system primarily focuses on centralized internship document submission, monitoring, and compliance tracking, while artificial intelligence features are integrated only as assistive tools for faculty review and report evaluation. The system will provide automated compliance tracking, real-time dashboards, structured workflows and document lifecycle management to support the entire lifecycle of an internship from pre-deployment through completion.

To further enhance system efficiency, selected intelligent features such as similarity detection and sentiment analysis will be incorporated as support tools to assist faculty in evaluating report submissions. These features are designed to complement, not replace, human decision-making. The AI-assisted features do not automate grading or approval decisions, as all evaluations remain under the supervision of faculty advisers and coordinators.

The system will be developed using an Agile Iterative Software Development Life Cycle (SDLC), enabling incremental delivery of modules aligned with the internship phases. The expected outcome will be a functional, scalable, and user-centered system that improves internship management, reduces administrative burden, and enhances transparency and compliance tracking at Saint Louis University.

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INTRODUCTION

1.1 Background of the Study

Universities play a critical role in advancing students' preparation for future careers, not only through academic instruction but also through experiential learning that integrates theoretical knowledge with real-world experience. Internship programs whereby students are allowed to work at a company for a limited time to gain practical work experience are one of the major methods employed by higher education institutions, and these programs, commonly called on-the-job training (OJT) in the Philippines, are frequently pointed out in the literature as the most effective single method. According to studies, internships that are carefully planned and well-organized are strongly linked to the increased level of employability of graduates, their readiness for a career, and the development of industry-relevant professional skills since such exposure to working environments, organizational culture, and sector-specific practices is very real and authentic for the students (Silva et al., 2018; Dawarka, 2024).

Internship programs in higher education in the Philippines are mainly regulated by the Student Internship Program in the Philippines (SIPP), introduced by the Commission on Higher Education (CHED) Memorandum Order No. 104, Series of 2017. SIPP dictates that all higher education institutions (HEIs) offering programs requiring practical experience must conduct internships that are well-organized, supervised, and comprehensively documented. The prescribed procedures indicate that each internship should be formally documented, institutionally supervised, and assessed through structured mechanisms, with the responsibilities of the HEI, Host Training Establishment (HTE), and the intern clearly identified (CHED, 2017). Working in accordance with the SIPP is not just a matter of choice but rather a regulatory obligation for institutional accreditation and program recognition.

At Saint Louis University (SLU), the School of Accountancy, Management, Computing and Information Studies (SAMCIS) oversees the internship programs for three degree programs under the Computing and Information Studies (CIS) department. Students in the Bachelor of Science in Information Technology (BSIT) program must complete a 600-hour internship in areas such as software development, network administration, information systems management, and technical support under the supervision of their mentors. The Bachelor of Science in Computer Science (BSCS) program requires a total of 240 hours of internship, primarily focused on providing students with exposure to research in computing, software engineering principles, algorithm development, and systems design. Bachelor of Multimedia Arts (BMMA) students need to complete a 500-hour internship in creative and technology-oriented environments, such as media production, digital content creation, animation, and other creative-technology sectors.

Across all three programs, the internship cycle is divided into three main stages: pre-deployment, active deployment, and post-deployment. The pre-deployment stage is the one that requires the most paperwork, since each student must obtain, submit, and have approved a total of 9 compliance documents before being officially recommended to the HTE. These requirements, outlined in Table 1, are the university's means of ensuring that the student is physically and legally compliant, properly placed within the organization, and made aware of the institution's policies before deployment.

Table 1: Pre-deployment Requirements for Student Interns

Requirement	***Description***
Medical Certificate	A legitimate medical certificate issued by a qualified physician confirming that the student is physically capable of engaging in workplace activities.
Psychological Clearance	A competent psychologist's certificate of psychological clearance attests to the student's emotional and mental readiness for a professional workplace.
Parent's Consent and Undertaking	Students who are under the legal age or who will be deployed outside of their home must complete a parental consent form.
Company Reply Form	A formal acceptance letter from HTE attesting to the student intern's deployment acceptance.
Data Privacy (DP) Orientation Certification	Evidence of completing the university's Data Privacy Orientation, which is mandated by the Philippine Data Privacy Act of 2012 (R.A. 10173).
Internship Orientation Completion Record	A document attesting to the student's participation in the required department-level internship orientation, covering rules, regulations, and student rights.
Government-Issued Company Permits	Copies of the HTE's Local Government Unit (LGU) Business Permit or SEC Registration attesting to the host organization's validity.
Work Plan	The department coordinator will review and approve a pre-approved document detailing the student's intended tasks, work schedule, internship goals, and expected contributions.
Memorandum of Agreement (MOA)	A legal agreement outlining the duties, responsibilities, and boundaries of each party between SLU and the HTE. Must be in place either before or throughout the orientation process.

Completing and obtaining approval for all pre-deployment requirements constitutes the institutional clearance of a student. It allows the student to begin formally counting their internship hours. Any missing or unapproved requirements at this stage must be addressed before the student is deployed. Weeks and months of accomplishment reports, attendance records, and tracks of hours spent must be submitted by students to the system during the active deployment phase. The post-deployment phase ends with the submission of a final technical report and the issuance of completion documents. The system should facilitate organized communication and monitoring among students, faculty advisers, department coordinators, and institutional administrators throughout all phases of this process.

Currently, internship document submission and compliance monitoring at SAMCIS primarily rely on Google Classroom, with manual processes such as spreadsheets, email, and physical document handling serving as supplements. Since Google Classroom was originally designed for managing general coursework, it mainly offers basic file uploading and assignment submission features. Nevertheless, it is not capable of handling the multi-stage, multi-stakeholder compliance workflow required by SIPP-internships.

Table 2 outlines the current workflow at each internship phase and highlights the operational constraints arising from reliance on this general-purpose platform.

Table 2: Current Internship Workflow at SLU SAMCIS and Identified Limitations

Phase	***Current Process at SAMCIS***	***Identified Limitation***
Pre-Deployment	Students submit compliance documents as separate Google Classroom assignments. Each student's submission column is manually reviewed by faculty.	No view of consolidated compliance. There isn't a system-imposed gate to stop deployment before every item is authorized. MOA records are handled by email and distinct files.
Active Deployment	Students upload weekly reports to Google Classroom as files. Students manually record their hours and attendance, and informal verification is done.	No methodical review process. There is no way to identify duplicate report entries. Hours are calculated by hand and are subject to error. No tracking of attendance deviations.
Post-Deployment	Google Classroom is used for submitting final reports and completing documentation. Letters of recommendation and certificates are drafted manually.	Document generation is not automated. Formatting discrepancies between outputs. There are no system-generated traceability reference codes.

Department Level Coordination	Coordinators manually gather compliance data from each Google Classroom submission. Spreadsheets and reminders are used to track MOA expiration.	No dashboard for the entire department, in real time. At-risk students are hard to spot. MOA expiration alerts are not automatic.

Not having a dedicated system for monitoring compliance with SIPP-mandated internships leads to a disjointed workflow, where important processes such as managing MOAs, calculating hours, and creating documents are handled outside the integrated platform. This scattering of processes results in increased administrative work, reduced record traceability, and severely restricted visibility that all stakeholders need to successfully manage the entire internship cycle (Laudon & Laudon, 2020; Alshibly, 2014).

A review of related literature and existing systems reveals that the problems encountered by SLU SAMCIS are not exclusive to the institution and that web-based platforms tailored for internship management have successfully addressed similar issues at other institutions.

Alsolais (2022) illustrated, via the case of a health college student internship portal launched on the web, that centralized online platforms dramatically reduce administrative burden, improve document tracking, and enhance communication among students, supervisors, and institutional coordinators compared to manual processes or email. Jaafar et al. (2017) developed a web-based system for internship monitoring and supervision that could assign supervisors in an organized manner, track schedules, and report progress, thereby demonstrating the superiority of purpose-built platforms over general-purpose tools. Nurfaizi and Hindarto (2023) developed a web-based student internship attendance application that not only simplified the calculation of hours but also streamlined attendance tracking, thereby minimizing errors and increasing the reliability of records of internship completion.

John (2023) researched the hurdles to adopting an electronic document and records management system in a public sector setting and noted that a centralized storage location, access controls, and audit logging are essential components for an institution to be compliant. For managing educational documents using AI, Russell and Norvig (2021) and Madanchian and Taherdoost (2023) describe the growing use of natural language processing tools, such as sentiment analysis and text similarity detection, to analyze written materials and identify patterns in educational documents. Tan et al. (2023) and Munezero et al. (2013) go further in showing that sentiment analysis is quite useful in educational environments, not only for assessing students' engagement but also for their welfare. The above findings lend support to the idea of embedding AI-assisted capabilities within IDSMS as mere aids to faculty evaluation, rather than taking over human judgment.

Table 3 provides a side-by-side comparison of the systems studied and the specific issues the proposed IDSMS aims to address.

Table 3: Comparative Summary of Related Systems

System / Study	***Institution / Source***	***Key Features***	***Gap Addressed by IDSMSCIS***
Web-Based Student Internship Portal	Health colleges, Saudi Arabia (Alsolais, 2022)	Online submission of documents, assigning supervisors, and monitoring of status	No AI-assisted report review, no MOA monitoring, and no computation of hours
Internship Monitoring and Supervising Web-Based System	Office of International Cooperation Universitas Nasional (Jaafar et.al, 2017)	Monitoring schedules, assigning supervisors, and reporting progress	No creation of documents or enforcement of pre-deployment checklists
Student Internship Attendance Application System	Indonesian University (Nurfaizi & Hindarto, 2023)	Automated hour computation and web-based attendance tracking	Attendance only; no report management or compliance monitoring
Electronic Document and Records Management System (ERDMS)	Public sector, St. Vincent & the Grenadines (John, 2023)	Access control, audit logging, and centralized document storage	Not internship-specific; neither dashboard monitoring nor workflow enforcement
Google Classroom (Current SLU Setup)	Saint Louis University, SAMCIS	Submission of assignments, uploading of files, and simple communication	No MOA management, no hours calculation, no compliance monitoring, and no organized review process

Upon reviewing related systems and examining the current workflow at SLU SAMCIS, a clear and consistent pattern emerges: general-purpose platforms like Google Classroom are insufficient to manage the multi-stage, documentation-heavy compliance requirements of SIPP-mandated internship programs. While existing purpose-built systems can handle specific aspects of internship management, such as attendance tracking, document submission, or supervisor assignment, none bring together the full spectrum of compliance requirements for the pre-deployment, active deployment, and post-deployment phases in a single platform aligned with Philippine SIPP guidelines.

The new Internship Document Submission and Monitoring System for CIS (IDSMS-CIS) will be proposed to address this issue. MOA lifecycle management, attendance and hours computation, structured report submission and review, pre-deployment checklist tracking, and automated document generation will all be centralized in a single web-based platform by IDSMSCIS, which will provide BSIT, BSCS, and BMMA programs of SAMCIS with a compliance and monitoring infrastructure built on the SIPP framework. The assistive AI features, in particular, similarity detection of weekly reports and sentiment analysis of journal entries, are incorporated as supplementary tools to support, not replace, faculty advisers' judgment in evaluating student submissions. The decision to build IDSMS will be based on proven institutional needs and the effectiveness of purpose-built internship management systems, as the existing literature indicates.

1.1.1 Current Workflow vs. Proposed Workflow

The figure below demonstrates the difference between the current internship management workflow at SLU SAMCIS and the proposed IDSMS-CIS workflow. In the current workflow, students upload compliance documents into individual Google Classroom assignment bins, faculty manually verify each submission, and any corrections or clarifications are communicated informally via private comments, Google Chat, or email. The department coordinator then collates submission statuses across multiple Classroom entries using spreadsheets, and final reports and summaries are compiled manually without any centralized system support. This process leads to siloed tracking, inconsistent turnaround times for feedback, and substantial administrative burden on both faculty and coordinators.

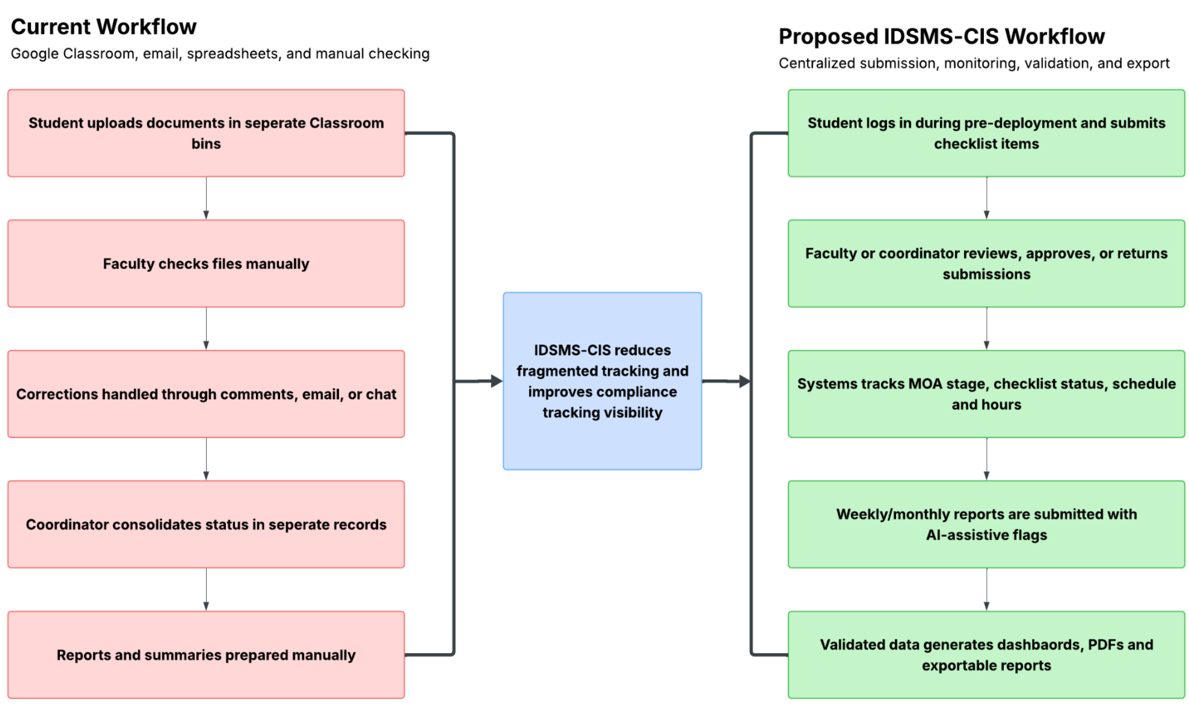

Figure 1: Current and Proposed Internship Document Workflow

These limitations are addressed by centralization in the proposed IDSMS-CIS workflow, which includes the following: During the pre-deployment phase, students log in to the system and submit checklist items through structured forms, which faculty advisers and department coordinators review, approve, or return in the platform, with all actions recorded and traceable. During the active deployment phase, the system automatically tracks MOA status, checklist completion, schedule adherence, and hours rendered. Weekly and monthly reports are submitted through structured forms with AI-assistive flags that surface potential quality concerns to faculty before review, and validated data is used to generate dashboards, PDF documents, and exportable compliance reports, eliminating the need for manual consolidation.

1.1.2 Business Process Flowchart

The business process flowchart shown below describes the end-to-end internship lifecycle as it is currently practiced at SLU SAMCIS from the search for a host training establishment (HTE) to the end of the active deployment phase. When a student requests an internship placement with an HTE or company, once a company is identified and agrees to accept the student, the work plan is submitted to the department head for approval. If the work plan is rejected, the student revises and resubmits until it is approved.

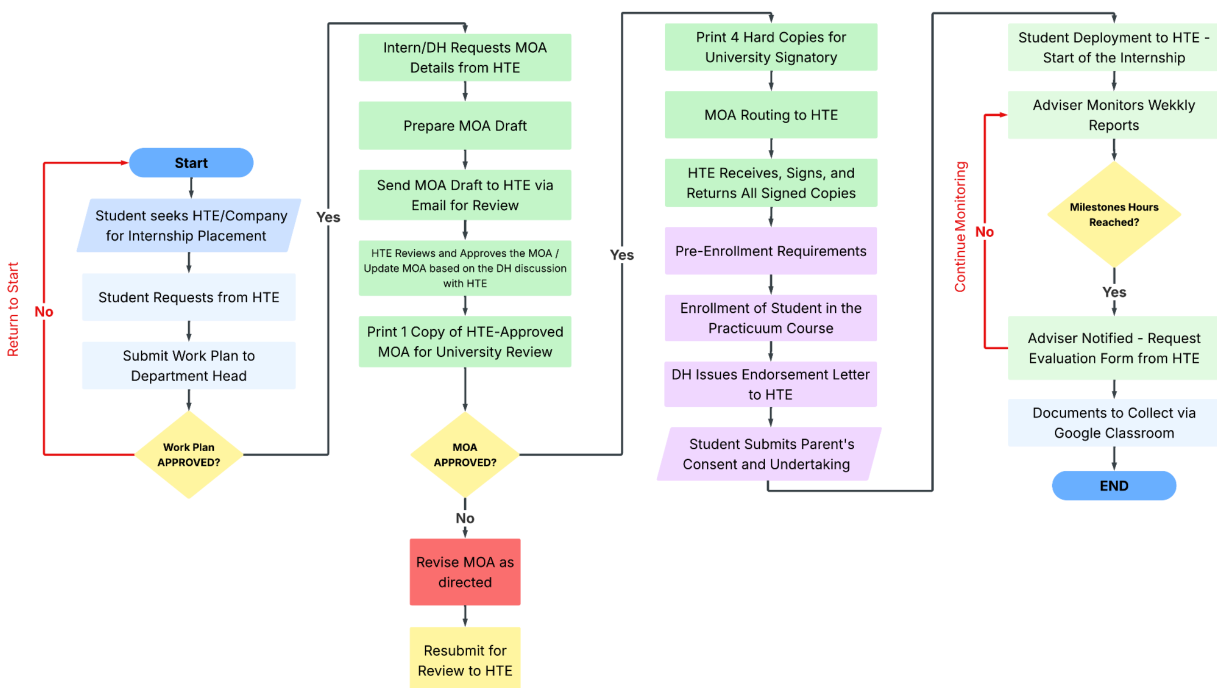

Figure 2: Internship Flowchart

After the work plan is approved, the MOA preparation commences. A draft is prepared and submitted via email to the HTE for review. The HTE reviews the draft and either approves or proposes revisions to the MOA. If revisions are needed, the draft is updated and resubmitted. Once the HTE approves the MOA, a copy is printed and sent to the university for review and final signing; four hard copies are printed and routed back to the HTE for their signatures. Once the MOA is fully signed, the student fulfills the remaining pre-enrollment requirements, such as enrolling in the practicum course, obtaining an endorsement letter from the department head, and submitting the parental consent and undertaking form to the HTE.

After these requirements are met, the student is deployed to the HTE, and the active internship phase commences, at which time the faculty adviser monitors the student via weekly accomplishment reports submitted through Google Classroom, checks that milestone hours have been met (if not, monitoring continues), and when the milestone hours are confirmed, requests an evaluation form from the HTE, and gathers any remaining completion documents through Google Classroom, thus completing the documented internship process.

1.2 Statement of the Problem

Students taking Bachelor of Science in Information Technology (BSIT), Bachelor of Science in Computer Science (BSCS), and Bachelor of Multimedia Arts (BMMA) under the School of Accountancy, Management, Computing and Information Studies (SAMCIS) of SLU have to go through structured internship programs from 240 to 500 hours with compliance requirements during pre-deployment, active deployment, and post-deployment phases.

The data from our interview with student interns suggests that the current internship documentation system still relies primarily on Google Classroom, emails, private comments, Google Chat, and manual checks. All three interns disclosed in their interviews that the weekly accomplishment reports are submitted through Google Classroom or other similar online submission channels. However, those very interviews illustrate that the existing arrangement is incapable of providing thorough monitoring, deadline control, requirement validation, revision tracking, or consolidated reporting. Not meeting these expectations has led to significant problems with submission compliance, document completeness, review turnaround, and report generation.

A Memorandum of Agreement (MOA) is an important step in the pre-deployment phase. One of the three intern respondents (100%) expressed frustration with the MOA processing as a significant constraint on their deployment, noting that students could officially begin only after the MOA was signed. The same intern shared that the MOA was sent via email, and there was a back-and-forth arrangement between the company and SLU when agreement details or NDA-related changes were necessary. Therefore, MOA processing is not merely paperwork; it is also a deployment checkpoint that requires status updates, validity checks, change tracking, responsible persons, and approval history.

Almost half of the interns interviewed (66.7%) reported incidents in which a weekly report was submitted late or could not be uploaded because the submission bin had already closed. So, in these cases, students had to resort to private comments, Google Chat, or email to explain the delay or share the missed file. This clearly indicates that late submission is not yet a formal procedure, but rather that issues are circumvented through informal workarounds.

Incomplete or incorrect submissions are common mistakes as well. Two of the three interviewed interns (66.7%) identified errors, including missing signatures, incorrect dates, mismatched hour counts, and reports requiring revision and resubmission. One intern even described a situation in which a single mistake affected several previous reports and required multiple corrected submissions. Another said that without a revised report submitted within the allowed timeframe, the weekly report could be considered invalid. That being said, these examples indicate that students sometimes miss very specific requirements. At the same time, faculty advisers have to manually identify, comment on, and even track missing or incorrect information.

Interviews further show that report reviews are neither immediate nor always conducted. One intern revealed that faculty review might take as little as 2 days or as long as a week, while the other said it could take 1 to 2 weeks to check. So, the latest review turnaround time ranges from 2 days to 2 weeks, depending on the faculty adviser, submission volume, and the condition of the report. Since feedback is communicated mostly through private comments, students probably do not have a consolidated view of which papers are accepted, rejected, revised, or still pending.

This manual process also literally increases the workload for faculty advisers and coordinators. Based on the reported workflow, each student submits weekly reports, monthly reports or journals, and final summary documents. For a typical 10-week internship period, one student may submit at least 10 weekly reports, excluding pre-deployment documents, monthly reports, attendance records, evaluation forms, certificates, and final summary reports. For every section handled by a coordinator, this creates a multiplying volume of files that must be opened, reviewed, commented on, signed, and monitored for completion. Since there is no live compliance dashboard at present, coordinators need to spend time consolidating submission status and identifying students with missing, late, or incomplete documents.

Another stage for report generation following deployment remains manual. Two of the three respondents (66.7%) mentioned that students themselves prepare or summarize final reports by using templates provided by the school. One of the interns mentioned that there was no definite, centralized record or summary of OJT performance kept for future reference. Other research and systems have shown that these issues are not limited to SLU. Web-based internship portals and monitoring systems have been developed to solve problems in document submission, attendance tracking, supervisor coordination, progress monitoring, and report management. To illustrate, Alsolais (2022) highlighted the importance of internship portals to reduce administrative burdens and improve member communication. Jaafar et al. (2017) developed a web-based system for monitoring and supervising internships, including supervisor assignment, schedule monitoring, and progress reporting.

Nurfaizi and Hindarto (2023) focused on web-based attendance monitoring and hour computation. In contrast, Dewi and Hanifah (2024) reported that their web-based internship monitoring information system improved the management of attendance, daily activities, and final report submission, with user testing indicating a satisfaction score of 87.53%. The above works show that there is a need for a specialized Internship Document Submission and Monitoring System for CIS (IDSMS-CIS), as general-purpose tools such as Google Classroom are not designed to manage the internship compliance cycle fully.

Thus, the biggest challenge tackled by this research is that there is no centralized, quantifiable, and well-organized system for submitting internship documents that also tracks student compliance, helps to cut down on late and incomplete submissions, facilitates faculty review, automates the process of tracking hours and reports, and gives coordinators instant access to the fulfillment of internship requirements.

1.2.1 General Problem:

The current internship management system at Saint Louis University lacks centralized compliance monitoring, leading to the use of Google Classroom and manual monitoring. How can the internship document submission, monitoring, and compliance tracking at Saint Louis University be improved through a centralized, AI-assisted system that complements Google Classroom?

1.2.2 Specific Problems:

1. How can pre-deployment requirements such as medical certificates, psychological clearance, parental consent, and company reply forms be efficiently tracked and managed within a centralized system?
2. How can the management and monitoring of Memorandum of Agreement (MOA) and company-related information be organized and made more accessible to authorized advisers?
3. How can weekly accomplishment reports, including attendance records (overtime, undertime, and absences), be effectively submitted, validated, and reviewed within a structured workflow?
4. How can the computation of total hours rendered, remaining hours, and expected internship completion dates be automated to reduce manual workload and improve accuracy?
5. How can real-time visibility of student compliance and submission status be provided to students, faculty advisers, and coordinators?
6. How can document generation, tracking, and record management be improved to ensure consistency, completeness, and traceability?

7. How can artificial intelligence be utilized to assist in validating documents, analyzing report quality, and supporting the evaluation process?

1.3 Objectives of the Study

1.3.1 General Objective

To develop a centralized Internship Document Submission and Monitoring System for IT, CS, and MMA programs at Saint Louis University.

1.3.2 Specific Objectives

In line with the Agile Iterative Software Development Life Cycle used for this project, the following measurable objectives are identified:

1. Requirements Analysis - To identify and document the functional and non-functional requirements of IDSMS-CIS through stakeholder interviews, document review, and workflow analysis of the current internship management process.
2. System Design - To produce the system architecture, database schema, use case diagrams, workflow diagrams, business process flowchart, and user interface prototypes for the four user roles: Student Intern, Faculty Adviser, Department Coordinator, and Super Admin.
3. System Development - To build the system modules for authentication and role management, pre-deployment checklist tracking, company and MOA management, attendance and schedule monitoring, weekly and monthly report workflows, document generation, dashboards, notifications, exports, and audit logging.
4. AI Integration - To integrate assistive AI functions for weekly report similarity detection and sentiment analysis, with results displayed only as faculty review indicators and not as automated grading or approval decisions.
5. System Testing and Evaluation - To verify and evaluate the completed system through unit testing, integration testing, performance checks, and user acceptance testing with students, faculty advisers, and coordinators based on functionality, usability, reliability, security, and performance criteria.

1.4 Scope of the Project

IDSMS-CIS accommodates three SAMCIS computer programs: BSIT (600 hrs), BSCS (240 hrs), and BMMA (500 hrs), through all internship phases. Pre-deployment includes nine compliance requirements per student, company profile management, and MOA lifecycle tracking with automated expiry reminders. While on deployment, the system is capable of managing an organized submission and review track of weekly and monthly reports, automatically calculating hours from approved schedules, and keeping track of attendance. The final technical report feature will be available only after hourly requirements have been satisfied. Role-based dashboards allow students, faculty advisers, and coordinators to see the submission status, document completeness, and hours rendered live, with semester-level archiving to assist in accreditation. Faculty members are given advisory indicators by AI-assisted tools, only similarity detection and sentiment analysis; however, no other aspects of grading or approvals are automated. The system is compatible with desktop and mobile browsers; a company supervisor portal, native mobile app, and grading functions are specifically excluded.

1.5 Significance of the Study

IDSMS-CIS benefits various groups within the Saint Louis University internship system. Student interns get a clear, centralized view of their compliance requirements, deadlines, and faculty feedback. This clarity reduces confusion and late submissions. Faculty advisers receive a single dashboard that brings together all student submissions. It includes AI-assisted indicators for similarity and sentiment, helping them focus their reviews effectively. The coordination office has visibility into compliance across departments, along with reports that can be exported and alerts for MOA that support accreditation reviews and institutional audits. For the wider research community, this study shows how to apply NLP-based sentiment analysis and vector-similarity detection in compliance settings. It offers a model that other higher education programs can replicate.

1.6 For Future Researchers:

This study serves as a foundation for future improvements in internship management systems by demonstrating how a centralized, AI-assisted platform can address the limitations of general-purpose tools like Google Classroom in managing internship compliance. Future researchers may build on this work by integrating more advanced AI techniques, expanding the platform into mobile applications, and incorporating data analytics to track trends and support better decision-making in university internship programs.

Chapter 2

METHODOLOGY

This chapter presents the methodology used to develop the Internship Document Submission and Monitoring System for CIS (IDSMS-CIS) for the IT, CS, and MMA programs under SAMCIS at Saint Louis University. The study follows an Adapted Agile Iterative Software Development Life Cycle (SDLC) composed of six phases: Requirements Analysis, System Design, System Development, Testing, Deployment, and Evaluation. This methodology was selected because the system comprises multiple interconnected modules spanning distinct phases of the internship lifecycle, making incremental delivery and progressive testing more suitable than a single-pass, sequential approach.

Standard Agile frameworks, such as Scrum, assume recurring stakeholder meetings at the end of each sprint to validate delivered increments and update the product backlog. In the capstone setting, this assumption does not hold: institutional stakeholders, including faculty advisers, coordinators, and student interns, are not available for scheduled sprint-review sessions throughout the development period, and panelists review the project only at the formal defense. To address this constraint, the methodology adapts Agile in two specific ways.

First, requirements were comprehensively gathered in the Requirements Analysis phase through structured interviews, document review, and adviser consultation, resulting in a System Requirements Specification that served as the fixed baseline for all design and development decisions. This approach front-loads stakeholder input rather than distributing it across sprints, ensuring that the development team works from validated requirements without needing ongoing stakeholder availability.

Second, sprint validation is conducted through adviser-guided milestone checkpoints at the end of each major sprint, rather than formal sprint reviews with the full stakeholder group. These checkpoints are scheduled when a module or group of related modules is functional and reviewable, allowing targeted feedback without assuming weekly availability. This adaptation is consistent with established practice in academic software development, where the adviser serves as the primary proxy for institutional requirements between formal panel evaluations (Kumar & Saxena, 2024).

2.1 Requirements Analysis

The requirements analysis phase was the primary channel for stakeholder input in this project. Because full stakeholder availability is limited to the defense period, all functional and non-functional requirements were identified and validated before development began. The research team collected information through three methods.

Interviews with student interns produced the foundational data for the problem statement. Three student interns currently enrolled in or recently completing their internship programs were interviewed using a structured guide covering document submission practices, late submission experiences, MOA processing, faculty review turnaround, attendance recording, and final report preparation. The interviews confirmed measurable problem indicators: 66.7% of respondents experienced late submissions or closed submission bins requiring informal workarounds; 66.7% reported incomplete or incorrect submissions requiring revision; and all respondents used Google Classroom or email as their primary submission channel despite these limitations.

Document review and workflow analysis involved examining the existing internship forms, submission checklists, report templates, MOA records, endorsement letter formats, and institutional guidelines provided by the project adviser. This analysis identified the required data fields, approval sequences, document output formats, and compliance checkpoints that the system must support.

Adviser consultation provided the operational depth needed to accurately define the system's scope. The project adviser, who oversees internship coordination, contributed detailed knowledge of the existing workflow, including MOA processing stages, faculty review practices, checklist structure, hour computation methods, schedule change handling, and deviation report requirements. This consultation was the primary source for functional requirements that could not be derived solely from student interviews.

The outputs of this phase were documented in a System Requirements Specification (SRS) that served as the binding reference for all subsequent design and development work. Any feature not traceable to the SRS was excluded from the development scope.

2.2 Development Framework

The development framework organizes the confirmed requirements from the SRS into a structured four-sprint Agile plan. Before development begins, the research team converts the SRS into a product backlog containing user stories, acceptance criteria, priority levels, and module dependencies. Each backlog item is assigned to a sprint and mapped to a specific study objective.

2.2.1 Sprint structure

Each sprint follows a fixed sequence: planning, design refinement, module development, unit and integration testing, and internal review. At the end of each sprint, the completed modules are demonstrated to the project adviser during a scheduled milestone checkpoint. Feedback from these checkpoints is incorporated into the next sprint before new modules are started. This mechanism replaces the stakeholder sprint review of standard Scrum while preserving the iterative and incremental character of Agile development.

2.2.2 Backlog management

The product backlog is maintained by the project leader and reviewed at the start of each sprint. Items are reprioritized based on module dependencies and any feedback received from the adviser checkpoint. Changes to requirements that emerge during development are assessed against the SRS; if they are within scope, they are added to the appropriate sprint backlog. Changes outside the scope are deferred and documented as recommendations for future development.

2.2.3 Sprint plan overview

Development is organized into four sprints. Sprint 1 establishes authentication and user management; Sprint 2 delivers the pre-deployment and company management modules; Sprint 3 covers active deployment workflows, including attendance, reporting, and faculty review; and Sprint 4 integrates AI-assisted indicators, role dashboards, notifications, document generation, and semester archiving. Full details for each sprint, including focus areas, modules, and deliverables, are provided in Table 6.

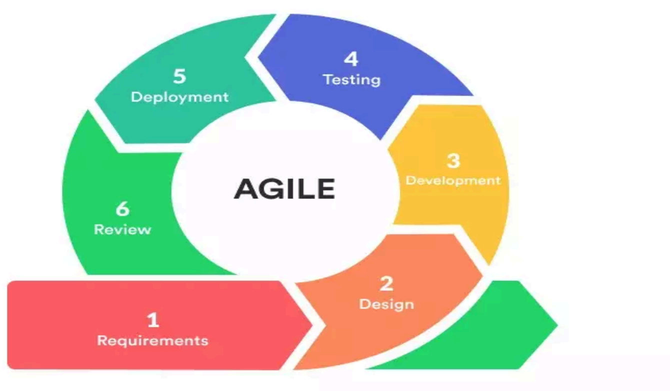

Figure 3: Agile Iterative Approach

2.3 Mapping of Objectives to Methodology

Table 4 demonstrates how the methodology achieves the different study objectives. Each objective will be linked to an SDLC phase, activity, and expected output through this mapping exercise.

Table 4: Mapping of the Objectives to Methodology Phases

Objective	Phase	Activity/Output	How Objective is Achieved
Requirements Analysis	Phase 1: Requirements Analysis	Stakeholder interviews, document review, workflow observation	A System needs Specification (SRS), which is the definitive baseline for all subsequent stages, identifies and documents both functional and non-functional needs.
System Design	Phase 2: Design	System architecture, database schema, use case diagrams, UI prototypes	An architecture, a 16-table database schema, and Figma prototypes are used to model all five user roles and three internship periods. This ensures that every design choice is based on a documented requirement.
System Development	Phase 3: Development	Modular monolith in Next.js; modules for auth, MOA, checklists, attendance, reports, document generation	Every sprint produces a functioning, tested increment. Sprint limits ensure progressive compliance with SIPP regulations by directly aligning with internship lifecycle phases.
AI Integration	Phase 3: Development	Sentiment analysis and similarity detection via Gemini API	AI elements are incorporated into the report review workflow as server-side service functions (aiService.ts) to support faculty evaluation rather than to replace it.

System Testing & Evaluation	Phase 4: Testing & Evaluation	Unit/integration testing (Jest), UAT with faculty and students, performance testing	Unit and integration testing per sprint verifies individual service functions and module interactions. User acceptance testing with students, faculty advisers, and coordinators, conducted through guided task scenarios, confirms that each workflow meets its acceptance criteria. Performance and reliability checks validate response times, PDF generation, and concurrent user handling. Results are documented and compared against criteria before each sprint is considered complete.

2.4 System Design

The validated requirements are transformed into a technical, functional, and visual design for IDSMS-CIS during the design phase. The design phase produces five primary design outputs: system architecture, database schema, use case diagrams, workflow diagrams, and user interface prototypes. These deliverables serve as the definitive reference for the application's build and testing.

2.4.1 System Architecture

The system architecture will follow a modular monolith pattern within a single Next.js application. The major components will include the client interface layer, application and service layer, authentication and role-based access control layer, data layer, secure file storage layer, AI assistance layer, notification layer, and audit logging layer. This architecture aligns with the project scope because it supports multiple modules while avoiding the deployment complexity of a full microservices architecture.

The AI assistance layer will be limited to two weekly report support functions: similarity detection and sentiment analysis. Similarity detection will compare new accomplishment entries with previous entries from the same student, while sentiment analysis will classify report reflections to help faculty notice possible morale or workload concerns. AI outputs will be stored as advisory indicators and will not control approval, grading, or completion decisions.

2.4.2 Database Schema

The database architecture specifies the sixteen objects necessary for the operation of the system: user, student profile, class group, semester, company, MOA record, pre-deployment checklist item, work plan, weekly report, daily report entry, deviation report, generated document, notification, audit log, holiday calendar, and system configuration. All primary keys are based on universally unique identifiers (UUIDs). Soft delete support is implemented in all elements of the schema using nullable fields for deletedAt or archivedAt to maintain full historical data for institutional audit purposes.

2.4.3 Use Case and Workflow Diagram

A use case diagram is created for Student Intern, Faculty Adviser, Department Coordinator, and Super Administrator. Workflow Diagrams outline the complete end-to-end business process for pre-deployment document submission, MOA management, attendance validation, weekly report review, and AI-assisted faculty evaluation. These diagrams represent the primary reference for implementing role-based access control (RBAC) and back-end business logic.

2.4.4 User Interface Prototypes

Figma uses a variety of prototypes to design interfaces for all user roles. Prototypes include dashboards, submission forms, checklist management screens, attendance logs, report review pages, document generation interfaces, and administrative configuration panels. Design prototypes should be validated against the SRS to ensure that all functional requirements are met via the interface before beginning development.

2.4.5 Technology Stack

The technology stack will be selected based on requirements identified during discussions, the academic context of a fourth-year capstone project, and alignment with modern, well-documented technologies with strong community support. The following table summarizes the primary technologies that will be adopted across each layer of the system.

Table 5: Recommended Technology Stack

Layer	***Technology and Role***
Framework	Next.js 14 (App Router) — full-stack React framework handling routing, server-side rendering, and API routes

Language	TypeScript — type safety across frontend and backend, reducing runtime errors in team development
Styling	Tailwind CSS — utility-first CSS framework providing a consistent design system
Database	PostgreSQL 15 via Supabase — relational database for all structured internship data
ORM	Prisma — type-safe database access with schema-driven migrations
Authentication	NextAuth.js — session management, credential provider, JWT strategy, and RBAC hooks
File Storage	Supabase Storage — secure bucket-based file storage with scoped access via signed URLs
Email	Resend + React Email — transactional email API with JSX-based template design
PDF Generation	Puppeteer (server-side) — headless Chrome for consistent server-rendered PDF output
AI: Embeddings	Gemini 2.5 Flash text-embedding-3-small — vector embeddings for similarity detection between report entries
AI: Analysis	Gemini 2.5 Flash— sentiment analysis
Vector Storage	pgvector (Supabase extension) — stores and queries embedding vectors within PostgreSQL
Testing	Jest + Testing Library (unit/integration); Cypress (end-to-end)
Deployment	Vercel — serverless Next.js deployment with automatic preview deployments and edge CDN
CI/CD	GitHub Actions — automated test runs on pull requests; auto-deploy to Vercel on merge to main

2.5 System Development

The development phase will transform the approved design into a working system through four Agile sprints. Development will begin with the core user and access-control foundation, followed by internship compliance modules, report and attendance workflows, and the final dashboard, export, document-generation, notification, and assistive AI components.

2.5.1 Development Process Overview

Following the Next.js App Router convention, the feature-based, modular folder structure includes a self-contained feature module for each feature, such as authentication, reports, companies, AI services, notifications, documents, and audit logging, with components, server actions, API route handlers, and TypeScript type definitions. Business logic is never placed directly in a page or route handler file; instead, it is extracted into a separate service file within the /src/lib/services/ directory (e.g., reportService.ts, checklistService.ts, aiService.ts) to support unit testing and maintain code quality.

2.5.2 Sprint-by-sprint Development Plan

Table 6 details the focus area, modules and features that will be developed and the expected deliverable for each of the four development sprints.

Table 6: Sprint Development Plan

Sprint	***Focus Area***	***Modules/Features***	***Sprint Deliverable***
1	Authentication & User Management	User registration and CSV/Excel bulk import; role-based access control (RBAC) for four roles (Student, Faculty, Coordinator, Super Admin); session management with auto-expiry; first-login password enforcement; student profile setup (program, class group, semester linkage); OTP/email MFA support; and immutable audit logging for login and role events	Working login system with RBAC middleware; role-gated navigation and dashboards for all four user roles; student profile screens; first-login password change flow
2	Company, MOA & Pre-Deployment	Company profile registration with autocomplete and duplicate detection; MOA record management (PDF/Drive upload, validity period, programs covered, stage tracking: Drafting → Active → Expired); automated MOA expiry email and in-app alerts; pre-deployment checklist tracking for all nine required documents per student (with approve/return/comment per item); work plan submission using institutional template; endorsement letter gating (blocked until all nine checklist items approved)	Complete pre-deployment workflow from student login to endorsement readiness; company/MOA module with expiry alerts; checklist submission and faculty approval workflows; endorsement letter lock enforcement

3	Deployment Phase - Attendance & Reports	Fixed work schedule configuration post-work-plan approval; mid-internship schedule change handling; calendar-based attendance recording populated with Philippine national/regional holidays and company non-working days; deviation reports (absence, overtime, undertime) with proof attachments and faculty validation; automated hour computation (total rendered, remaining, projected completion); weekly report submission via auto-populated structured forms with copy-paste detection flagging; monthly report aggregation from four weekly reports; faculty review workflow (Approve, Return, Regard, Disregard) with email notification triggers	Complete deployment-phase workflow; calendar-based attendance log; automated hours progress dashboard; structured weekly/monthly report submission and faculty review screens
4	AI Integration, Documents & Dashboard	Similarity detection uses the Gemini text-embedding model with pgvector, setting a threshold of 0.70 for flagging. Sentiment analysis is performed by Gemini 2.5 Flash, which categorizes outputs as Positive, Neutral, or Negative, along with a confidence score. Cached AI results are available without live API calls at page load. PDF generation is standardized through Puppeteer for endorsement letters, weekly and monthly reports, and final report cover pages, with system-generated reference codes. Compliance monitoring dashboards track per-student hours, checklist percentages, submission status, and anomaly flags for all four roles. A notification system with 12 triggers is managed via Resend. The system features semester archiving, complete audit logging, and compliance export to Excel or CSV.	Fully integrated system with AI-assisted faculty review; automated PDF document generation with reference codes; real-time compliance dashboards for all roles; notification system; semester archive module; system ready for UAT and evaluation

2.5.3 Database Implementation

The database structure will be composed of tables that will hold data on users, student profiles, class groups, semesters, companies, MOA records, pre-deployment checklist items, work plans, weekly reports, daily report entries, deviation reports, generated documents, notifications, audit logs, holiday calendars, and system configuration. UUIDs will be consistently used for all primary keys. In addition, every table included creation and update timestamps to track changes. Soft deletes will be implemented by adding a nullable deletion timestamp field, so that no records will actually be deleted from the system.

2.5.4 AI Module Implementation

The AI module will be implemented as a set of server-side service functions. Similarity detection will use Gemini's text embedding model to generate vector representations of daily report descriptions, which will then be compared against the student's previous report embeddings stored in PostgreSQL using the pgvector extension. Reports with a similarity score exceeding 0.70 will be automatically flagged. Sentiment analysis will be performed using Gemini 2.5 Flash, with a structured prompt requesting classification of the report text as Positive, Neutral, or Negative, along with a confidence score and brief reasoning. All AI results will be stored in the database after computation and served from cache on subsequent requests, ensuring that no live AI API calls are required during page load.

2.5.5 Notification System Implementation

The notification system will be implemented as a centralized dispatcher that will route all email and in-app notifications. Automated notifications covered twelve trigger events, including account creation, password reset, report submission, report approval, report return, submission deadline reminders at 48 and 24 hours, endorsement letter generation, MOA expiry warnings, checklist item status changes, and work plan approval. All notification intervals will be configurable through the Super Admin interface and will not be hardcoded in the application logic.

2.5.6 Development Diagram

In Figure 4, the system development phase will be executed in four Agile sprint increments, starting with the approved requirements, system design outputs, repository setup, coding standards, environment configuration, and database connection. Each sprint will deliver a working module that will be integrated with the shared backend services and tested by regression testing before the next sprint starts.

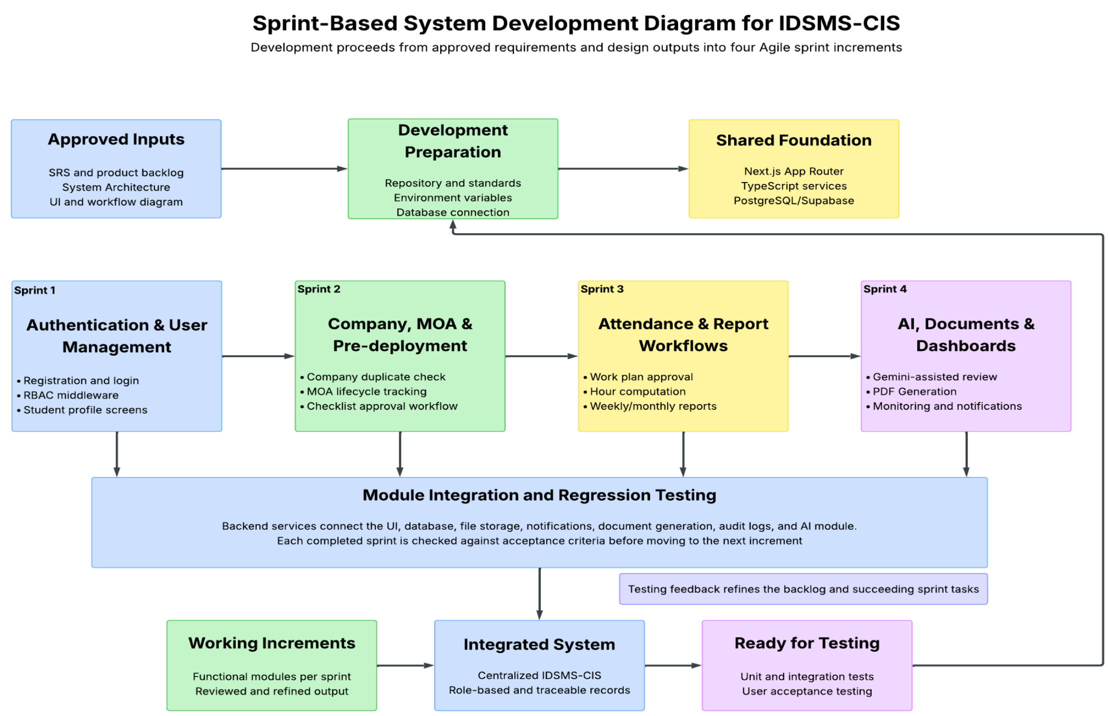

Figure 4: Sprint-Based Development Diagram for IDSMS-CIS

The diagram also indicates that testing feedback will be used to refine the product backlog and enhance subsequent sprint tasks, so authentication, company, and MOA management, checklist tracking, attendance, and report workflows, AI-assisted analysis, document generation, dashboards, and notifications will be developed in an orderly sequence that can be traced.

2.6 Testing

Testing will be conducted throughout the development process and at the end of each sprint. Unit testing will verify individual service functions, including checklist updates, hour computation, report validation, MOA status transitions, notification triggers, and AI result handling. Integration testing will verify interactions among modules, such as whether an approved work plan correctly affects weekly report schedules and hour computation.

User Acceptance Testing will be conducted with selected student interns, faculty advisers, and coordinators through guided task scenarios. Participants will test common workflows, including logging in, submitting requirements, returning a report, validating attendance, reviewing AI flags, generating an endorsement letter, and exporting a compliance summary. Feedback will be recorded and used for refinement.

Performance and reliability checks will examine page loading, dashboard response, PDF generation, export generation, and handling of simultaneous users. Testing results will be documented and compared against acceptance criteria before each sprint is considered complete.

2.7 Evaluation Framework

The system will be evaluated after the development sprints using technical, functional, and user-centered criteria. Technical evaluation will consider response time, PDF generation speed, export generation, secure access, and system reliability. Functional evaluation will determine whether the system correctly tracks checklist items, MOA stages, schedules, attendance, report statuses, rendered hours, AI indicators, and generated documents.

User evaluation will use a structured feedback form after guided testing sessions with students, faculty advisers, and coordinators. Criteria will include ease of use, clarity of interface, usefulness of dashboards, accuracy of status indicators, reduction of manual tracking, usefulness of export features, and overall satisfaction compared with the current Google Classroom-centered workflow.

The evaluation will also consider compliance and privacy. The system will be checked against SIPP-related documentation and monitoring needs, role-based access requirements, consent logging, secure file handling, and the limitation that AI indicators remain assistive rather than decision-making tools.

2.8 Ethical Considerations

As the IDSMS system processes highly confidential personal and academic information, such as medical certificates, psychological clearance documents, parental consent forms, and attendance records, data privacy and security will be considered the main pillars during the system's design. Access to information is controlled through role-based access control: a student can only see his/her own records, a faculty adviser can only see the information of the interns assigned to him/her, and coordinators can see more, but only within the scope of their department.

All communications between the client and the server are secure because they are encrypted using SSL. Every protected route requires the system to re-authenticate the user. Additionally, the personal documents uploaded to the system are stored securely, and unauthorized users cannot access them. The team has also considered the use of the system's AI capabilities; for example, sentiment analysis of journal entries is used only to support evaluation, and it is emphasized that students' performance is not determined solely by the AI.

Furthermore, consolidated consent is provided to interns for awareness of the use of AI to analyze the documents they submitted through the platform, in accordance with the law, specifically RA 10173, to protect the fundamental human right to privacy of communication while ensuring the free flow of information for innovation, growth, and development.

Chapter 3

PROPOSED SYSTEM DESIGN AND SPECIFICATIONS

This chapter presents the proposed outcomes of the development of the Internship Document Submission and Monitoring System (IDSMS) at Saint Louis University. The system will be successfully developed as a centralized, web-based platform that complements Google Classroom. The following sections describe the key outputs produced across all four development sprints and the extent to which the system's objectives will be met.

3.1 Functional and Non-Functional Requirements

3.1.1 Functional Requirements

The functional requirements of IDSMS specify the system operations and functionalities that should be recognized to effectively handle the internship compliance management issues identified. The desired features have been extracted from stakeholder interactions, analysis of relevant documents, and observation of the process. The requirements are organized according to the system's main functional units.

Table 7: Summary of Functional Requirements by Module

Module	Core Functional Requirements
User Management and Authentication	The system must accommodate four user types: Super Admin, Department Coordinator, Faculty Adviser, and Student Intern. Explicitly, each of these roles must have distinct access rights, controlled through role-based access control (RBAC), at both the user interface and API levels. Using institutional class lists, the system should enable bulk student registration via CSV or Excel file import. The system shall create default passwords based on a salted hash of the student ID number. It is mandatory to enforce password changes at first login. Furthermore, for very sensitive operations, the system should provide options for one-time password (OTP) or email-based multi-factor authentication. Aside from that, the system should support session management that automatically expires after a specified period of inactivity. Also, there must be an immutable audit log that records all login attempts, role changes, and security-sensitive actions.

Company Profile & MOA Management	The system should allow students and faculty to register host company profiles that include the company name, full address, work modality type (On-site, Work-from-Home, or Hybrid), supervisor or HR contact information, and student position title. When entering company names, the system should support autocomplete and duplicate-detection functionality. MOA records should be uploadable as PDF files or Google Drive links, with system-captured metadata on the MOA's validity period and the academic programs covered. The system should automatically identify MOAs approaching expiration and send email and in-app alerts to the internship coordinator. Government-issued business permits should be uploaded and verified by faculty or coordinators, after which the student is activated for the internship.
Pre-Deployment and Requirements Checklist	The system should track each of the following nine pre-deployment documents for each student, with a separate item for each, allowing faculty to approve or return each checklist item with written comments, and displaying the student's checklist completion percentage in real time: medical certificate, psychological clearance, parental consent and undertaking, company reply form, Data Privacy orientation certificate, internship orientation completion record, government-issued company permits, approved work plan, and Memorandum of Agreement. The system should prevent the generation of endorsement letters and progression to the deployment phase until all mandatory checklist items have been approved. Data Privacy Orientation and Internship Orientation should be tracked separately, and the system should log user consent to data collection and processing when the user registers for the orientation.
Work Plan Submission and Approval	Students should submit a work plan prior to the internship, based on the standardized template provided by the institution, and have it approved by the department coordinator, with the system verifying that the planned tasks match the student's academic program; if the company has its own work plan, the system should notify the student and request additional information be provided. Approved work plans must be used to develop the fixed work schedule, calculate the required hours, and determine the expected completion date of the internship; the system should automatically generate an endorsement letter upon approval of the work plan.

Attendance and Time Tracking	Upon approval of work plan, each student must set up their work schedule (daily hours and working days) in the system, and the system must populate a calendar-based schedule with Philippine national holidays, regional holidays, and company-specific non-working days that students can mark as applicable, with absences, undertime, and overtime reported via structured deviation report forms with date, reason, and supporting documentation, and all deviation reports must be validated by the faculty adviser prior to it affecting the hour computation for the student. The system must automatically compute the total hours rendered, required hours left, and projected completion date for each student based on validated attendance data. Required completion hours per program must be configurable via an admin interface, not hardcoded in the application.
Weekly and Monthly Report Submission	The system should generate weekly report forms based on each student's approved fixed work schedule and the semester calendar. The form should include daily attendance status, actual work hours, work accomplishments, total hours for the week (auto-calculated), running total hours, and hours remaining. Only the student should enter actual hours worked and daily accomplishments; copy-paste detection should identify accomplishment entries that are essentially identical to earlier submissions and prompt the student to revise them before submission. Reports should be due on Tuesday of the following week, and a note to the faculty adviser must be submitted before late submissions are allowed. The faculty should be able to approve or return reports via a structured review workflow and should receive email notifications when reports are approved or returned. Monthly reports should aggregate four weekly reports with the same structure and review workflow.
Final Technical Report	A technical report module must be made available to the student only after they reach a configurable hour threshold (e.g., 500 of 600 required hours for BSIT) and must contain sections for company profile summary, peer and supervisor evaluation scores, and proof of attachments (e.g., proof of attendance, certificates of completion, and certificates of participation).
Document Generation	The system should automatically create endorsement letters with the student name, ID number, company name and address,

	supervisor name and position, and endorsement date (using the standard letterhead format of the institution), and faculty should be able to create endorsement letters for all students in a class group using a batch PDF export function. The system should create PDF versions of weekly reports, monthly reports, endorsement letters, and final report cover pages, including a system-generated reference code and timestamp for traceability and accreditation, and PDF generation should occur only after the student has reviewed and confirmed the report contents in the system.
AI-Assisted Features	The system would need to include sentiment analysis of weekly report journal entries to gauge intern morale, work engagement, and potential welfare concerns, with aggregated sentiment trend indicators surfaced on the faculty dashboard. A similarity detection module would compare new accomplishment entries to the previous report embeddings using vector-based comparison and flag entries above a configurable similarity threshold before submission. Results from AI would be cached and served to avoid live API calls on page load.
Dashboard and Monitoring	Students, faculty advisers, and coordinators must be provided with role-specific dashboards, and the system must provide real-time per-student submission status, total hours rendered, pre-deployment checklist completion percentage, attendance anomaly flags, and company and modality breakdown, which must be auto-generated into weekly dashboard summary reports showing the number of reports submitted, validated, pending, and flagged for export to Excel for offline review and accreditation use. All historical data, per semester, must be retained to support institutional audit requirements, and the system must support an end-of-semester archiving process.
Notification System	Automated emails for account creation, password resets, report submission confirmation, report return with revision notes, upcoming submission deadline reminders at configurable intervals, endorsement letter generation, MOA expiry warnings, and pre-deployment checklist status changes should be sent. In-app status indicators should be displayed for returned reports, pending

	validations, upcoming deadlines, and incomplete pre-deployment requirements.

3.1.2 Non-Functional Requirements

Non-functional requirements characterize the quality attributes the system should have when deployed within the institutional context and will be specified by the development team based on institutional policy, national internship program guidelines, and the Philippine Data Privacy Act of 2012.

Table 8: Non-Functional Requirements Summary

Attribute	Key Requirements
Security	Role-based access control must be enforced across all system layers, with users able to access only data within their granted scope, all client-server communication must be encrypted with HTTPS/TLS 1.2 or greater, uploaded files must be stored in a secure, authenticated cloud storage backend and not be publicly available without a valid, scoped authentication token, file uploads must be validated server-side with MIME-type verification and be limited to safe file types (PDF, DOCX, PNG, JPG, XLSX, CSV), sensitive actions (role changes, bulk exports, PDF generation, and user deletion) should require re-authentication or MFA confirmation, there should be rate limiting on authentication endpoints to defend against brute-force attacks, and there should be an immutable audit log with all data access, modification, export, and deletion events, including the user identity, timestamp, and affected record.
Data Privacy and Compliance	The system should be compliant with the Philippine Data Privacy Act of 2012 (Republic Act No. 10173) and SLU's institutional data privacy policy, with student personal data, including health-related documents, psychological records, and parent consent forms, accessible only to personnel on a need-to-know basis and encrypted at rest, explicit user consent to collect and process data logged during account registration and the orientation flow, data minimization applied to only collect information necessary for internship management, system not expose student personal information in publicly accessible URLs, shared links, or unscoped

	API responses, and implement a data retention policy that flags records for institutional review or anonymization after the defined retention period, rather than automatic deletion, and allow users to request a copy of their stored personal data.
Usability	The system should have an easy-to-use interface that does not need much training time for students and faculty, an in-system onboarding guide or tooltip system, autocomplete, pre-filled defaults, inline validation messages to reduce input errors and manual data entry, and should be fully responsive and accessible on desktop and mobile browsers without needing a native mobile application, calendar and schedule interfaces should be able to differentiate working days, non-working holidays, and submission deadlines. Status indicators throughout the system should use consistent color-coded visual language (e.g., submitted, validated, returned, pending, and overdue states).
Reliability and Availability	The system should be online during academic working hours, with maintenance windows communicated to users at least 48 hours in advance; automated data backup procedures are required with a minimum frequency of daily backups; PDF generation should be consistent and accurate across different browsers and devices; and the system should handle failed submissions gracefully, retaining draft data and displaying a clear recovery message to the user.
Maintainability	The architecture should provide modular, service-oriented capabilities so that any component can be independently updated without compromising the overall solution's functionality. The AI and vector database layers should be physically and operationally distinct from the relational database layer, thus supporting independent scaling, model updates, and feature toggle functionality. Configuration parameters such as program completion hours, holiday calendars, semester dates, submission deadlines, and other time-sensitive information should be configurable through an administrative user interface (UI) rather than hardcoded into the application logic. All code must conform to documented coding standards, and unit, integration, and end-to-end testing will be developed with a minimum coverage requirement. All third-party integrations should be implemented as abstracted service adapters

	that allow replacing the underlying provider without refactoring the core solution.
Scalability	The system architecture should allow for the addition of more academic departments without necessitating structural changes to the database schema or application core, support multiple semesters, academic years, and departments, store full historical data without performance degradation, and handle bulk operations such as bulk student enrollment, bulk endorsement letter generation, bulk report export, and bulk checklist review. The AI feature layer should be designed for horizontal scaling to handle concurrent requests for sentiment analysis and similarity detection during peak submission periods.
Integration	The system needs to integrate with Google Classroom for the file submission and return platform, with the IDSMS web portal being the monitoring, validation, and compliance layer above it, integrate with the SLU institutional student portal for enrolled student lists, class codes, and ID numbers for auto-registration, integrate with an authenticated transactional email service for automated notifications, credential delivery, and OTP authentication, and all dashboard reports and checklist summaries must be exportable to Excel format (Microsoft Excel and Google Sheets) and integrate with a secured cloud storage solution for document uploads and MOA management.

3.2 Designing and Implementing of the System

This section discusses the design and implementation decisions for the Internship Document Submission and Monitoring System, including the architectural pattern, use case model, data architecture, technology stack, and prototype interface for all four development sprints.

3.2.1 System Architecture

IDSMS utilizes a Modular Monolith architecture pattern, where all functional modules, such as authentication, report management, company and MOA management, AI-assisted analysis, document generation, notifications, and audit logging, reside within a single Next.js application codebase, but are organized as independently testable, loosely coupled feature modules. This approach was chosen over a full microservices architecture because of the project's development constraints: it is being developed by a fourth-year capstone team, must still be deployable as a single project, and should not incur the operational overhead of managing multiple independently hosted services.

The architecture will be designed in six layers, each with defined responsibilities. At the top, the client layer will be composed of React components rendered by Next.js and served from Vercel's edge content delivery network, with users interacting with the system through browser-based interfaces on desktop and mobile, and real-time status updates delivered via server-sent events or polling, without WebSocket infrastructure. Under the client layer, the Next.js application layer handles page routing via the App Router, server-side rendering, server actions, and API route handlers. All business logic is extracted into dedicated TypeScript service modules in the /src/lib/services/ directory, including reportService.ts, checklistService.ts, companyService.ts, notificationService.ts, aiService.ts, documentService.ts, auditService.ts, and userService.ts. API route handlers and server action files never directly contain business logic; they delegate to these service modules.

An authentication layer, powered by NextAuth.js, will handle authentication and access control, and a Next.js middleware file at /src/middleware.ts intercepts every request before any page or API route, validating the session token with the NEXTAUTH_SECRET, enforcing role-based access control (RBAC) at the route level, and redirecting first-login users to change their system-generated default password. RBAC will be enforced both at the middleware to restrict access to a route and at the service layer to validate record-level ownership, preventing a student from accessing another student's report even if they build a valid API request.

The data layer includes a PostgreSQL 15 database hosted on Supabase that is accessed through Prisma ORM for all standard queries (Prisma provides type-safe database access and schema-driven migrations, so raw SQL is not in application code), except pgvector similarity search queries used by the AI module, which use Prisma \'s $queryRaw with bound parameters to query vector embeddings stored in the daily_report_entries table. The second layer of enforcement is at the database level through Supabase Row Level Security policies, which are independent of the application.

The storage layer will be implemented with Supabase Storage, organized into access-controlled buckets by purpose, with all uploaded files stored within them (pre-deployment documents, MOA PDFs, deviation report proofs, and generated output PDFs). Files are never served via public URLs; all file access is mediated by the application server, which generates time-limited signed URLs scoped to the requesting user's authorization level.

The AI service layer is the only component that will support optional external delegation. The server-side service functions in aiService.ts make two different calls to the Gemini API: text-embedding-3-small to generate 1536-dimensional vector embeddings for similarity detection, and Gemini 2.5 Flash for sentiment analysis. All AI computations are triggered asynchronously after report submission, and results are persisted in the database, so no live AI API call is needed while any page is being served.

The notification and email layer will route all email and in-app notifications through a central notificationService.ts dispatcher that invokes the Resend transactional email API using JSX-based React Email templates. This service handles 12 notification trigger events: account creation, report submission, report return, submission deadline reminders, MOA expiry warnings, checklist status changes, and scheduled deadline reminders. Vercel Cron Jobs dispatches these at configurable intervals.

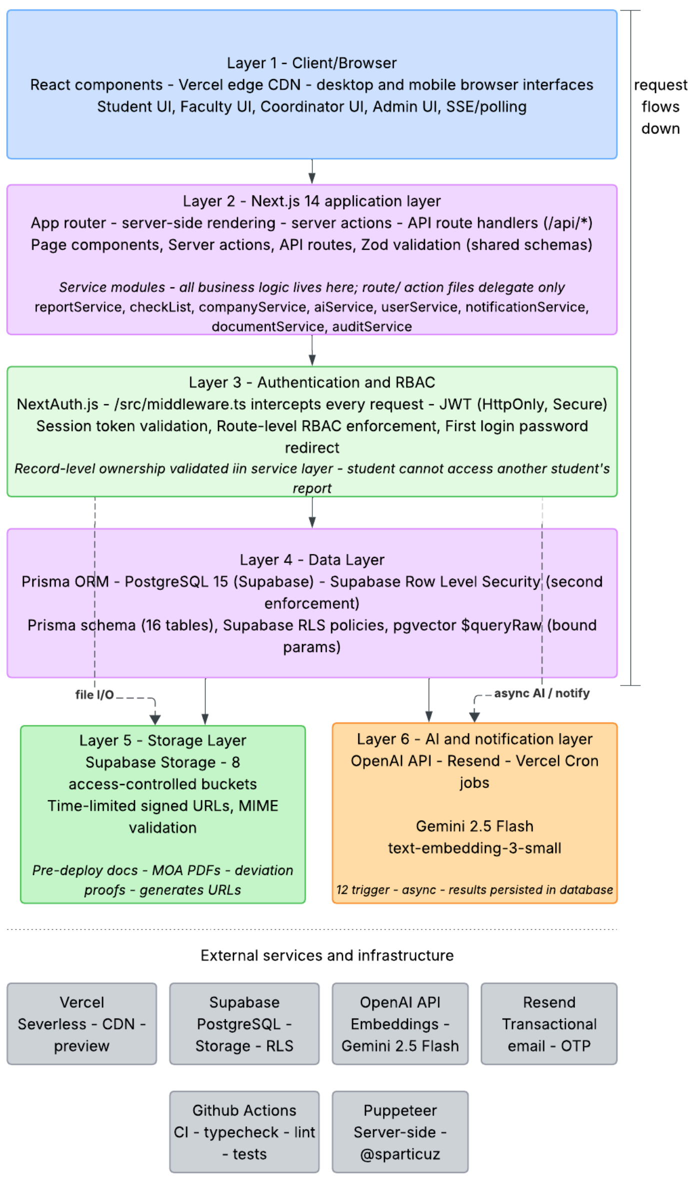

Figure 5: System Architecture Layer

3.2.2 Use Case Diagram

The use case diagram for the system will depict the interactions between the four defined user roles (Student Intern, Faculty Adviser, Department Coordinator, and Super Admin) and the functional capabilities of the system based on the Role-Based Access Control permission map and the full API route specification that is documented in the Technical Design Document, which maps every use case to an implemented system feature.

The Student Intern is the most active user, engaging in ten primary use cases across the entire internship lifecycle. In the pre-deployment phase, the student logs in, updates the system-generated first-login password, provides the nine pre-deployment checklist documents, registers the host company profile with duplicate detection, submits the work plan form using the institution-provided template, and configures and views the approved fixed work schedule after work plan approval. In active deployment, the student submits structured deviation reports for absences, overtime, and undertime; creates and submits weekly reports through an auto-populated form with copy-paste detection; creates and submits monthly reports aggregated from four weekly reports; and accesses and submits the final technical report once the program-specific hour threshold is met. In all phases, the student monitors hours of progress and consumption.

Figure 6: Use Case Diagram for Student Intern

The Faculty Adviser interacts with the following six primary use cases via the faculty dashboard: the adviser approves or returns individual pre-deployment checklist items with written comments, validates or rejects deviation reports before they impact the hour computation, reviews and validates or returns weekly reports through a structured workflow, views AI-generated similarity and sentiment flags for each reviewed report, monitors the compliance and submission status of all students in assigned class groups, and generates individual or bulk endorsement letters in PDF format. The faculty adviser also exports class compliance reports as Excel files for offline review and distribution.

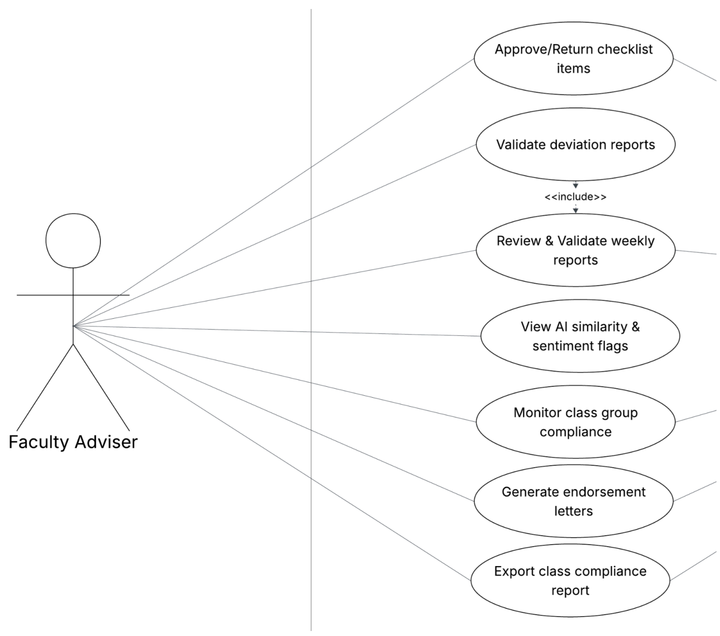

Figure 7: Use Case Diagram for Faculty Adviser

The Department Coordinator has higher level administrative use cases, such as approve or return student work plans, confirm planned tasks match enrolled program, manages Memorandum of Agreement records (upload MOA documents, track validity periods, receive automated expiry alerts), bulk student enrollment through CSV upload linked to official SLU class codes, approves or returns final technical reports upon completion, and shares access to the faculty-level monitoring, endorsement, and export use cases.

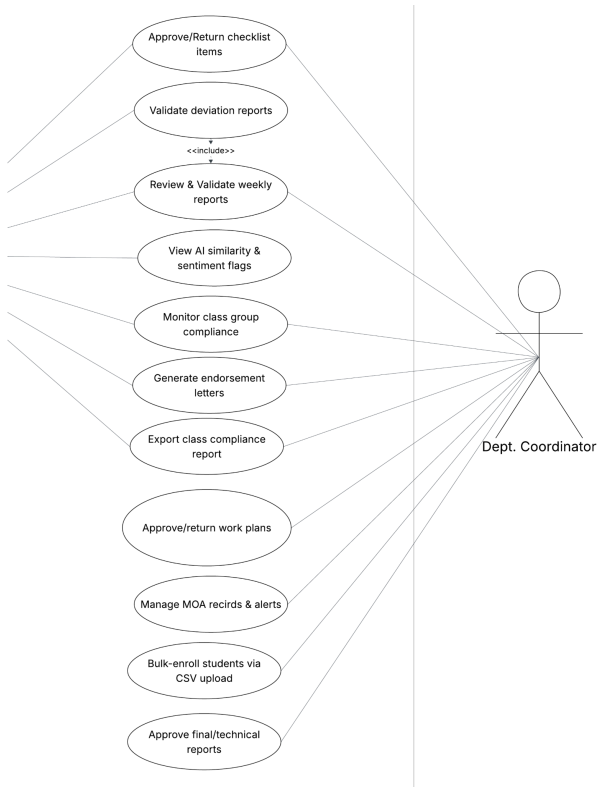

Figure 8: Use Case Diagram for Department Coordinator

The Super Admin role has use cases that no other role can access, including managing all user accounts, assigning and modifying roles, configuring system-wide settings such as required completion hours per program, similarity detection thresholds, deadline reminder intervals, and MOA expiry alert days through the admin interface; creating and archiving semester records; and accessing the immutable audit log that records all system activity.

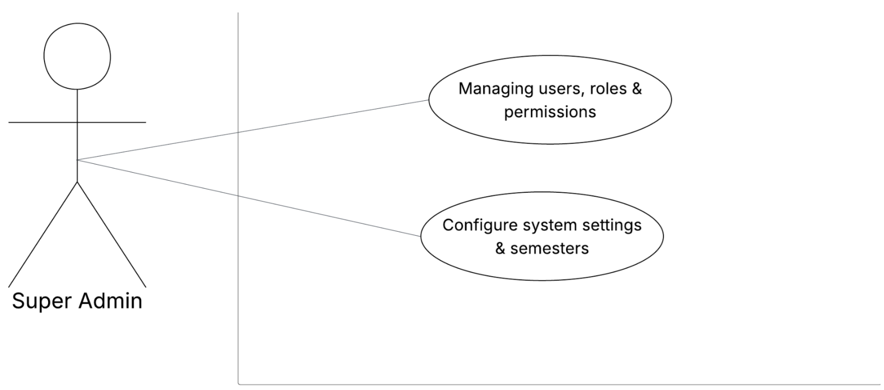

Figure 9: Use Case Diagram for Super Admin

The use case diagram also records two important relationships between use cases: an include relationship indicates that the Review and validate weekly reports use case includes the View AI similarity and sentiment flags use case, because the AI analysis results are always computed. Available as part of the report review workflow, and an extended relationship indicates that the similarity and sentiment flag display extends the base review use case, optionally when AI results have been computed.

3.2.3 Data Architecture

The users table will store account information for the four system roles: Student Intern, Faculty Adviser, Department Coordinator, and Super Admin. Student profile records link students to the program, class group, semester, company, schedule, required hours, and rendered hours. Class group and semester records will organize students by academic term and faculty assignment.

Company and MOA tables will store company profiles, work modality, supervisor or HR contact information, MOA status, validity dates, programs covered, and archival status. MOA stages will include Drafting, Pending, For HTE Review, For University Review, Approved/Active, Expiring/Expired, and Archived. Access to MOA records will be restricted to authorized Faculty Advisers and Department Coordinators.

Pre-deployment checklist records will be created per student and requirement type. Each checklist item will store status, submitted file reference, reviewer, review date, comments, and archival state. Checklist templates will use stable, unique identifiers so CSV uploads can add, ignore, or archive items without overlapping existing database records.

Work plan records will store approved tasks, schedule configuration, schedule-change history, and approval metadata. Attendance and deviation records will store daily attendance, absences, undertime, overtime, reasons, supporting proof, and validation status. Only validated records will affect rendered-hour computation.

Weekly and monthly report records will store report periods, daily entries, total hours, revision history, faculty action, and status. Faculty actions will include approval, return, regard, and disregard. AI-related fields will store similarity scores, similarity flags, sentiment labels, confidence indicators, and processing timestamps for advisory review only.

Generated document records will track endorsement letters, weekly and monthly report PDFs, export files, and reference codes. Notification records will log email and in-app messages. Audit logs will capture security-sensitive activities, including login attempts, role changes, checklist changes, exports, MOA updates, report decisions, and file access.

Holiday calendar and system configuration records will store program hours, semester dates, holidays, deadline intervals, MOA alert thresholds, similarity thresholds, final report unlock rules, session timeout, and other configurable values. These settings will allow the system to support changes without hardcoding operational rules. Figure 7 presents the database schema.

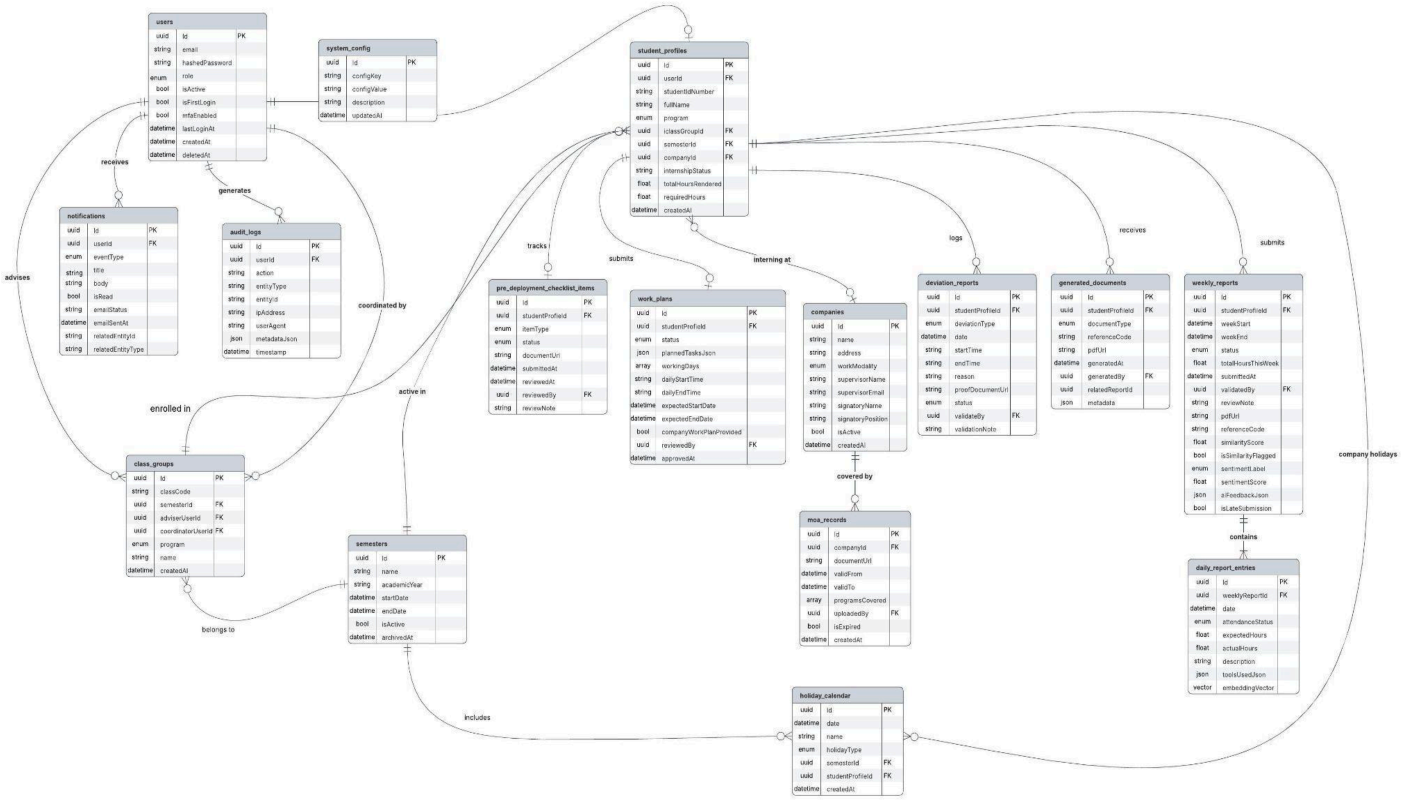

Figure 10: Database Structure

3.2.4 Technology Architecture

The proposed IDSMS adopts a modern, scalable, and modular technology architecture designed to support web-based access, real-time processing, and AI-assisted functionalities. The architecture follows a three-tier structure enhanced with service-based components, ensuring flexibility, maintainability, and scalability.

The system is composed of three primary layers: the presentation layer, the application layer, and the data layer, with an additional integration layer for an AI service.

Client Interface Layer (User Interaction)

The client interface layer provides access to the system for all identified user roles, namely students, faculty members, internship coordinators, and administrators. This layer implements the user interactions, including document submission, progress tracking, compliance checking, and dashboard visualization. Each user role is presented with a tailored interface that reflects their responsibilities within the internship lifecycle.

Application and Module Layer (Core System Logic)

The application layer serves as the core of the system. Each module corresponds to a specific function within the internship lifecycle. The Document Management Module handles the submission, storage, and organization of internship documents, including pre-deployment requirements, weekly reports, and final outputs. The Compliance Tracking Module monitors student progress by checking submitted documents against required deliverables. It generates real-time compliance status and alerts for missing or delayed submissions.

The Monitoring and Dashboard Module provides faculty members and coordinators with visual representations of student progress. It aggregates data from various stages of the internship lifecycle and presents them through dashboards, enabling efficient supervision. The Reporting and Evaluation Module supports the generation of summaries, evaluations, and completion reports. It ensures consistency in assessment and aligns with the completion phase.

All modules operate within a unified workflow engine that enforces role-based access control and process sequencing based on the internship lifecycle stages.

AI Processing Layer

This layer is a distinct component introduced to support the intelligent features. This layer enhances the system's ability to analyze submitted documents and assist in monitoring and evaluation.

It performs functions such as report validation, detection of incomplete or inconsistent entries, and identification of anomalies in submission patterns. These capabilities reduce the need for manual checks and improve the reliability of compliance monitoring. The AI layer interacts with the application layer by receiving document data for analysis and returning validation results, which are then reflected in the compliance tracking and dashboard modules.

Data Management Layer

This layer is responsible for storing and maintaining all system data. This includes user profiles, submitted documents, compliance records, evaluation results, and system logs. The database structure is derived from the entities and workflows, ensuring that each stage of the internship lifecycle is properly recorded and traceable. Relationships between users, documents, and compliance statuses are maintained to support accurate monitoring and reporting. Data integrity, security, and consistency are ensured through controlled access, validation mechanisms, and regular backup processes.

Lifecycle-Based Data Flow

The architecture supports the internship lifecycle defined by enabling structured data flow across stages.

During the pre-deployment stage, student data and initial requirements are captured and stored. During the active internship stage, ongoing submissions, such as weekly reports, are processed and analyzed. The monitoring stage utilizes compliance tracking and dashboards to evaluate progress, while the completion stage finalizes records and generates evaluation outputs. This lifecycle-driven flow ensures that all system components work cohesively and that data is consistently updated and accessible across modules.

Deployment Considerations

The system is implemented as a web-based application to ensure accessibility and compatibility with existing tools such as Google Classroom. It follows a client-server model, where users access the system through web browsers while processing and data storage are handled by server-side components.

3.2.5 Prototype

The prototype is an essential aspect of creating the IDSMS, providing a visual representation of the system's functionality before the entire team builds it. For the system's prototypes, Figma was used to define the core functionality layout and navigation flow, as well as to design pages and features for each of the limited number of primary users.

Features of the Student include Dashboard, Document Submission and Monitoring, Weekly Report Submission, Attendance Logs, Profile Management, and Account Settings. Features of Faculty Adviser include Dashboard, Monitoring Class Groups, Reviewing Reports, Pre-deployment Checklists, Generating Endorsement Letters, Profile Management, and Settings. Features of Department Compliance Coordinators include Dashboard, Manage Companies & MOA, Review Work Plans, Reports, Export, and Settings. Lastly, Features of Super Admin include Systems Settings, Role Assignments, Holiday Calendar Management, and Export Center, providing complete control of the system.

The design team can now visualize and create structured designs that are interactive, user-friendly, and aligned with the operational workflow for each user role. Table 9 consists of Figures 8 through 35 containing the UI for the various roles and the application's primary functional elements.

Table 9: User Interface Prototypes for IDSMS-CIS

Figure 11: IDSMS Logo

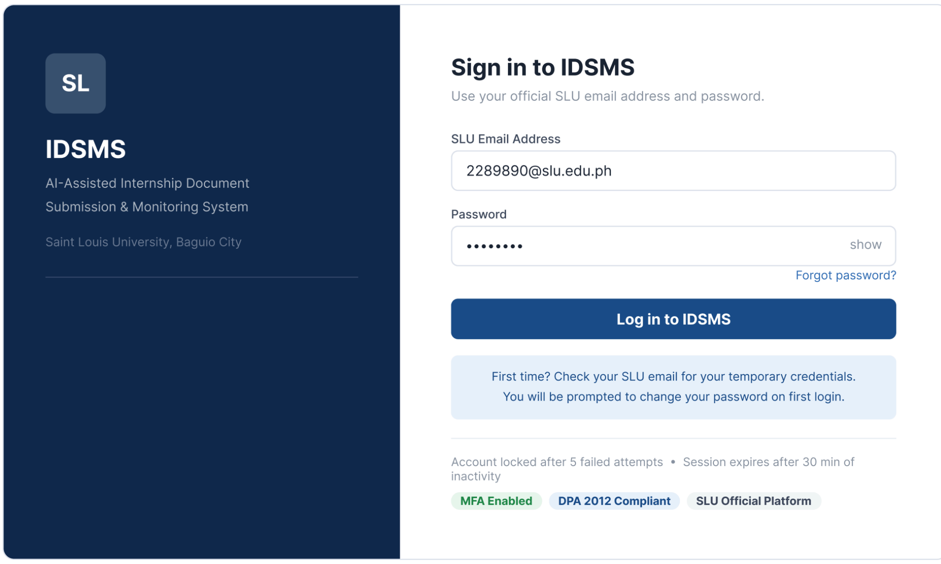

Figure 12: IDSMS Login Page

Description	The Login Page is the gateway for registered users to the IDSMS, and through it, users can log in using their official SLU institutional email address and password. The page supports MFA (Multi-Factor Authentication), includes a Forgot Password link, provides help for first-time logins, and displays security notices, such as an account being locked after 5 failed attempts and a session expiring after 30 minutes.
Pre-Condition	The user needs an existing IDSMS account in the system assigned by the System Administrator, a valid SLU institutional email address, and initial credentials (sent via email). The system needs to be live and open.
Process Required	1. The first step is that the user visits the IDSMS login page. 2. Next, the user fills in their SLU email address and password. 3. Then, the system checks the credentials by looking them up in the user database. 4. If the credentials are correct, then if MFA is turned on, the user is asked for their second piece of evidence of their identity. 5. Once everything is checked out, the user is taken to the page that has their information and functionality depending on their role. 6. Each unsuccessful login attempt adds to the lockout count; upon the 5th failure, the account is locked for some time.
Document/s produced	None

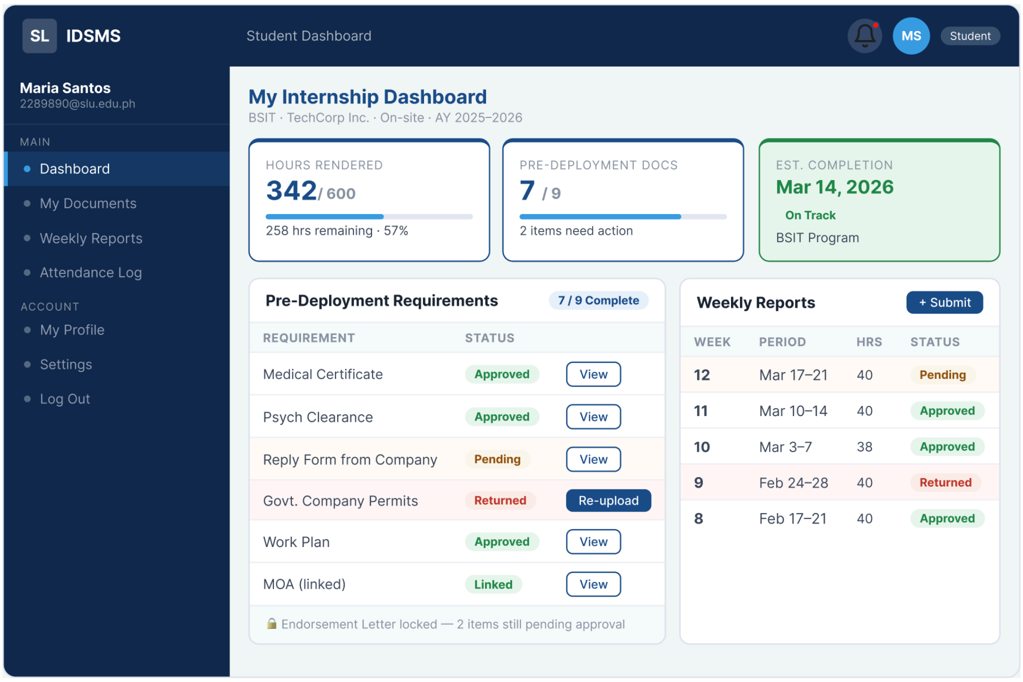

Figure 13: Student - Dashboard

Description	The Student Dashboard shows the student's main internship status at a glance. In addition to the hours rendered versus the required hours, the student's pre-deployment document completion, the estimated date of completion, a summary of pre-deployment requirements, individual statuses (Approved, Pending, Returned), and weekly report submission history with corresponding statuses are also presented.
Pre-Condition	The student must be logged in to the IDSMS system with the Student/Intern role. The student internship details (company, program, schedule) must be set up in the system by the coordinator or administrator beforehand.
Process Required	1. The student logs in and lands on the Dashboard. 2. The system fetches and shows the student's real-time data, such as hours rendered, document checklist completion, weekly report statuses, etc. 3. The student checks the list of pending items and takes steps to resolve any flagged documents by means of quick-action buttons (e.g., Re-upload, View). 4. The status of the Endorsement Letter is displayed as locked since no items have been approved yet.
Document/s produced	None

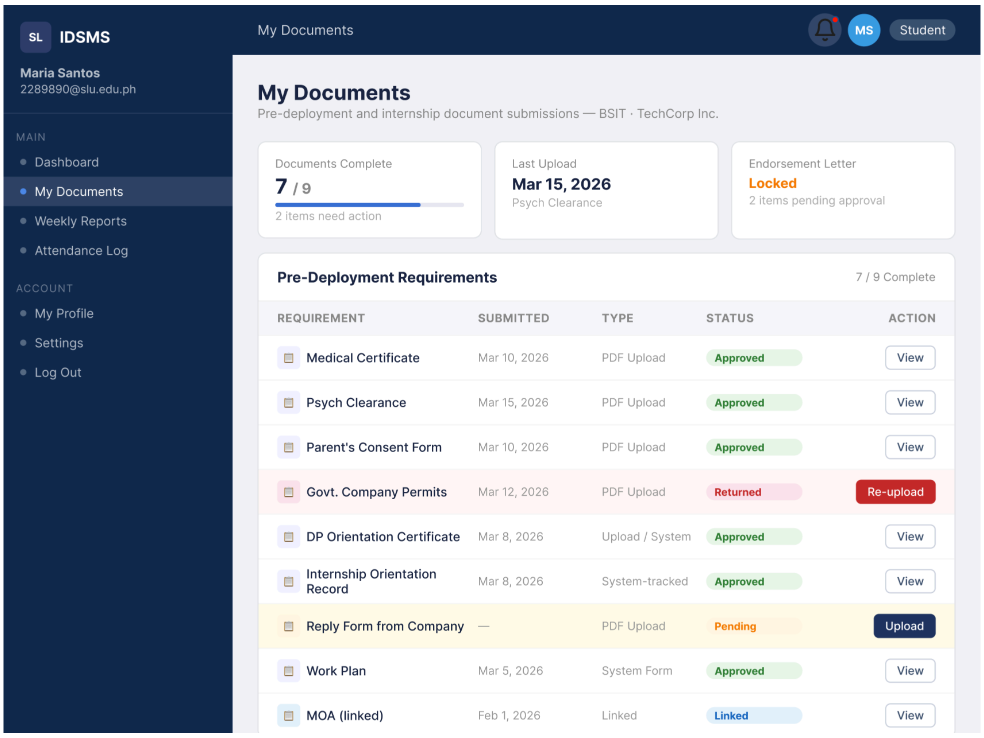

Figure 14: Student - My Documents page

Description	The My Documents page shows students the full range of document requirements for both the pre-deployment and internship phases. A single line in the list contains the document name, the date when it was submitted, the type of submission (PDF Upload, System Form, System-tracked, Linked), and the current status (Approved, Pending, Returned, Linked). Action buttons give students the option to View, Upload, or Re-upload documents. A summary header displays overall completion, last upload date, and endorsement letter lock status.
Pre-Condition	For the student to complete the documentation, he or she must be authenticated and enrolled in an active internship semester. The pre-deployment document requirements need to have been configured by the system coordinator for the student's program.
Process Required	1. The student goes to 'My Documents' in the sidebar. 2. The system retrieves all document requirements and their statuses from the database. 3. The student opens the documents to be uploaded for the ones with 'Pending' or 'Returned' status and clicks 'Upload' or 'Re-upload'. Then, the student submits the suitable file. 4. The uploaded files are stored and marked as ready for faculty review.

	5. The system refreshes the checklist progress and sends a notification to the faculty adviser.
Document/s produced	PDF documents uploaded (e.g., Medical Certificate, Psych Clearance, Government Company Permits, Reply Form from Company, Parent's Consent Form) are stored in the system as student submission records.

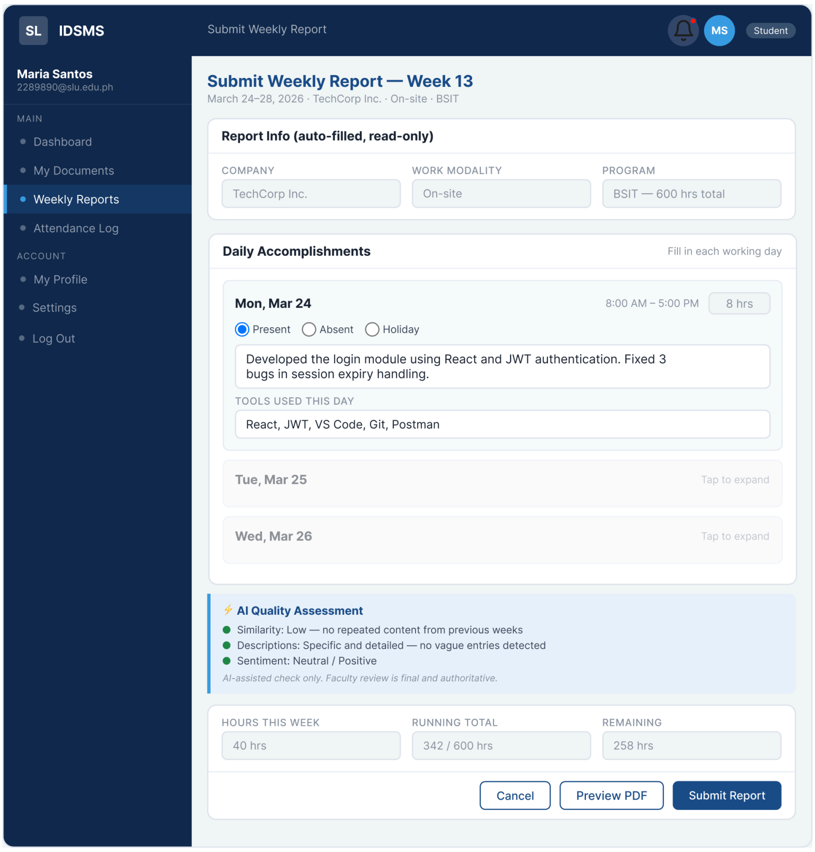

Figure 15: Student - Weekly Reports page

Description	The Student Weekly Reports section allows students to submit their weekly internship accomplishment reports. It will be a read-only form that shows the company and program details, which are prefilled. Students will be able to enter details of their daily accomplishments, including their attendance status (Present/Absent/Holiday), the number of hours worked, and the tools used each day. The AI Quality Assessment panel will check the report for similarity, specificity, and sentiment before submission. The form will reflect the total running hours and the remaining hours.
Pre-Condition	To submit a weekly report, a student must be logged in and assigned to an active internship. The current week's report period must be open for submission. Besides, students are not allowed to submit the current week's report if

	they have already done it.
Process Required	1. Students open the 'Weekly Reports' page and access this week's form. 2. Students complete daily activities and mark their attendance on each working day. 3. The AI tool analyzes the report for copying, ambiguity, and sentiment instantaneously. 4. Students go over the AI Quality Assessment responses. 5. Students select Preview PDF to check the output format, then hit Submit Report to send it. 6. The submitted report will be forwarded to the faculty adviser for checking.
Document/s produced	After submitting weekly reports, a PDF version will be automatically generated that summarizes students' daily accomplishments, attendance, and hours worked for that week.

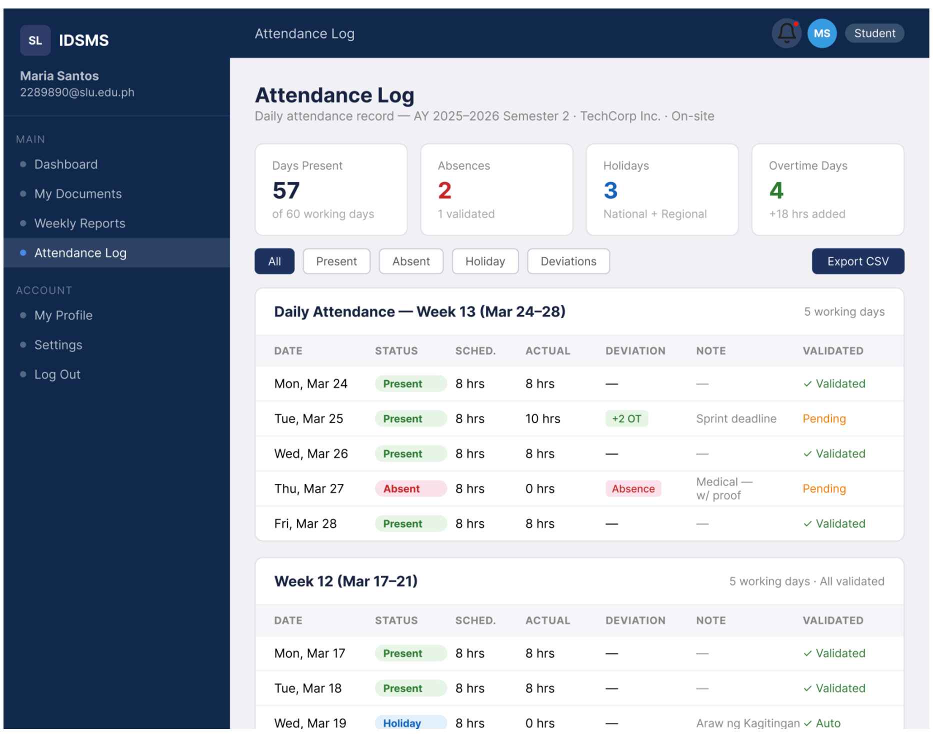

Figure 16: Student - Attendance Log page

Description	The Student Attendance Log breaks down a student's daily attendance for the entire semester. It summarizes attendance statistics (Days Present, Absences, Holidays, Overtime Days) and also provides detailed data for each week, including Date, Status, Scheduled Hours, Actual Hours, Deviation, Notes, and Validation status. Filter tabs enable viewing by All, Present, Absent, Holiday, or Deviations. Users can also export CSV for a download of their data.

Pre-Condition	Students must be logged in. Attendance data is pulled automatically from weekly report submissions and faculty adviser validation. At least one weekly report needs to have been submitted and validated.
Process Required	1. The student selects 'Attendance Log' from the sidebar. 2. The system collects attendance data from all submitted weekly reports. 3. Students can filter records by status (Present, Absent, Holiday, Deviations). 4. Deviations (e.g., overtime, absences) show up with notes as pending faculty validation. 5. Students are allowed to download the log as a CSV file for their own use.
Document/s produced	A downloadable CSV containing a student's attendance records for the semester.

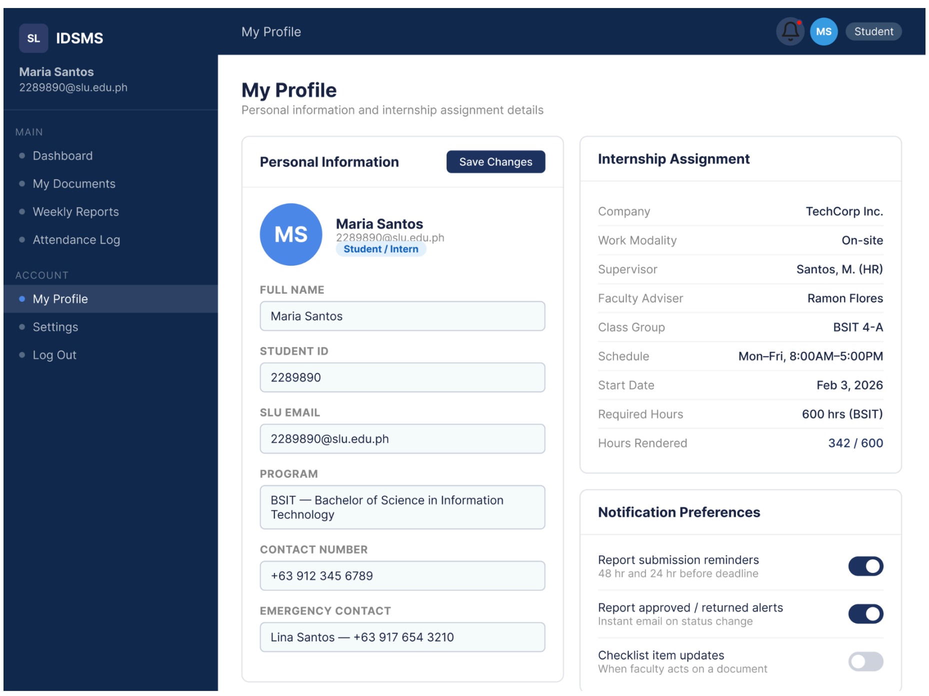

Figure 17: Student - My Profile page

Description	The Student My Profile page is where the student can view their personal information and internship assignment details. The personal information part contains fields such as full name, student ID, SLU email, program, contact number, and emergency contact. The internship assignment section displays the information like the company, work modality, supervisor, faculty adviser, class group, schedule, start date, required hours, and hours rendered. Also, students can configure their notification preferences for receiving report submission reminders, approval/return alerts, and checklist item updates.

Pre-Condition	First, the student has to be authenticated. Second, the system coordinator must set the internship assignment details before the student can see them.
Process Required	1. The student goes to 'My Profile' from the Account section. 2. Students may change editable personal fields (contact number, emergency contact). 3. Students press 'Save Changes' to save the updates. 4. Students can turn notification preferences on/off. 5. Internship assignment fields are non-editable and controlled by the coordinator.
Document/s produced	None

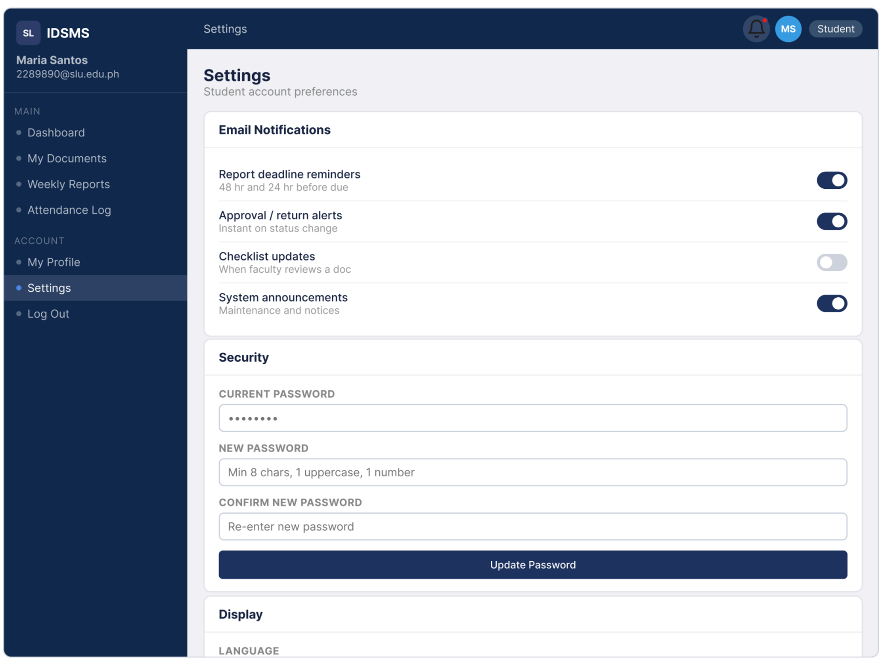

Figure 18: Student - Settings

Description	The Student Settings page is the place where the student can change their account settings, like turning on/off email notifications (reminders for report deadlines, alerts for approval/return, updates of checklists, announcement of the system), security settings (a password change with the minimum password complexity requirements), and display preferences (language selection).
Pre-Condition	The student needs to have an active IDSMS account to be able to login.
Process Required	1. The student goes to the sidebar and clicks on 'Settings'.

	2. Students can change email notification preferences as they want. 3. The student needs to provide the current password, new password, and confirmation password whenever they want to change passwords. 4. The system checks if the password meets the complexity requirements (at least 8 characters, 1 uppercase letter, 1 number). 5. Students can press 'Update Password' to make the changes effective. 6. The display language preference is changed immediately after the user selects the new language.
Document/s produced	None

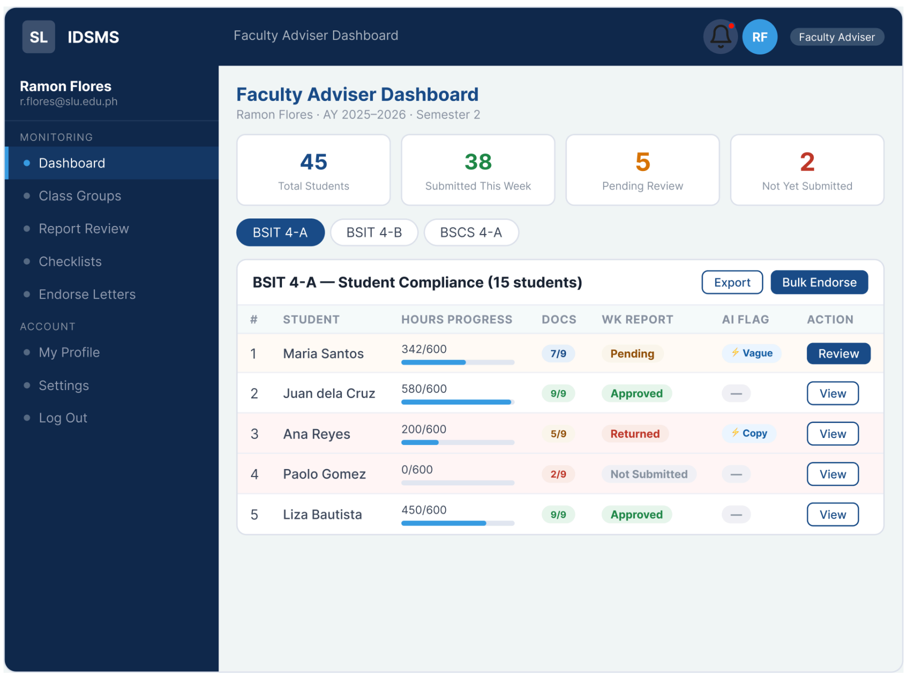

Figure 19: Faculty - Dashboard

Description	The Faculty Adviser Dashboard shows a summary of the compliance status of all students across the class groups assigned to the faculty adviser. It indicates the total number of students, No of reports submitted this week, reports pending review, and reports not yet submitted. The different class groups' view tabs consist of individual student rows showing progress bars for hours, number of completed documents, weekly report status, AI flags (Vague, Copy), and action buttons (Review, View). Bulk Endorse and Export are the next functions.
Pre-Condition	The faculty adviser needs to be logged in. The system administrator or coordinator must assign the class groups to the faculty adviser.

	Additionally, students must be enrolled in the current semester.
Process Required	1. Faculty logs in and is redirected to the Faculty Adviser Dashboard. 2. System fetches and displays real-time compliance data for all students assigned to the faculty. 3. Faculty clicks on a class group tab to display the list of students in that group. 4. The faculty selects 'Review' to examine a particular student's weekly report. 5. Faculty may 'Bulk Endorse' to approve several students eligible at one time. 6. Faculty may export the compliance list of class groups.
Document/s produced	Comprehensive downloadable compliance report (CSV/Excel) for each class group. Endorsement letters are created from the Endorse Letters module.

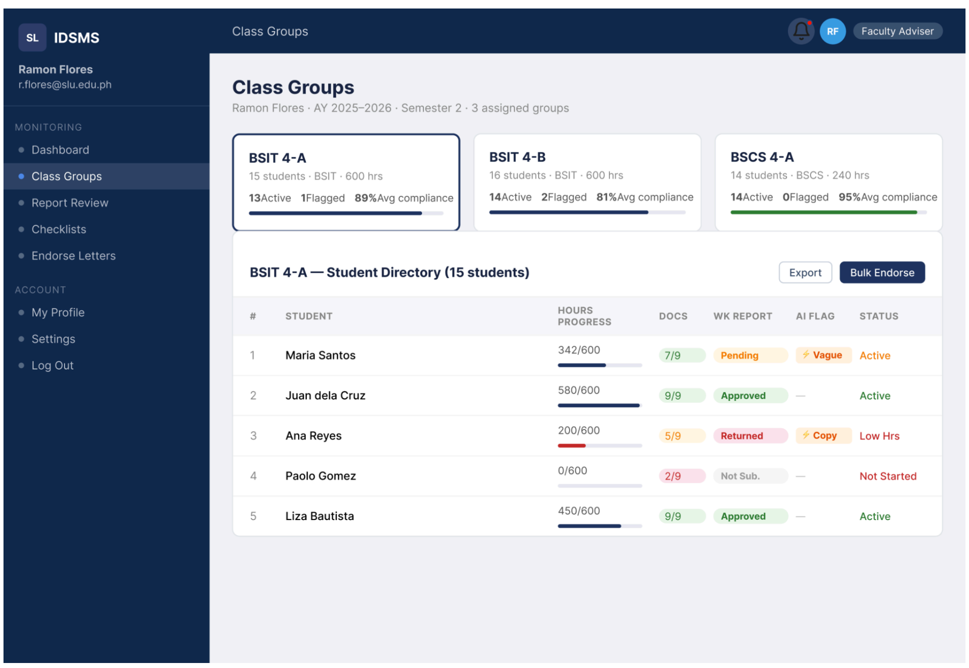

Figure 20: Faculty - Class Groups page

Description	The Class Groups page lists all class groups assigned to the faculty adviser during the semester. Each group card indicates the number of students, program, hours required, active students, flagged count, and the average compliance percentage. By clicking a group, one can see a table listing students with their hours progress, document count, weekly report status, AI flags, and student status labels such as Active, Low Hrs, and Not Started.
Pre-Condition	Faculty adviser login is required. Class groups

	should have been created and assigned by the department coordinator or system administrator.
Process Required	1. Faculty locates 'Class Groups' via the sidebar. 2. The system shows all the assigned class groups with summary statistics. 3. The faculty chooses a group for the student directory. 4. Faculty 'Bulk Endorse' allows processing of endorsement letters for eligible students in bulk. 5. Export option provides a download of students for the selected group.
Document/s produced	Student directory list exportable by class group (CSV/Excel format).

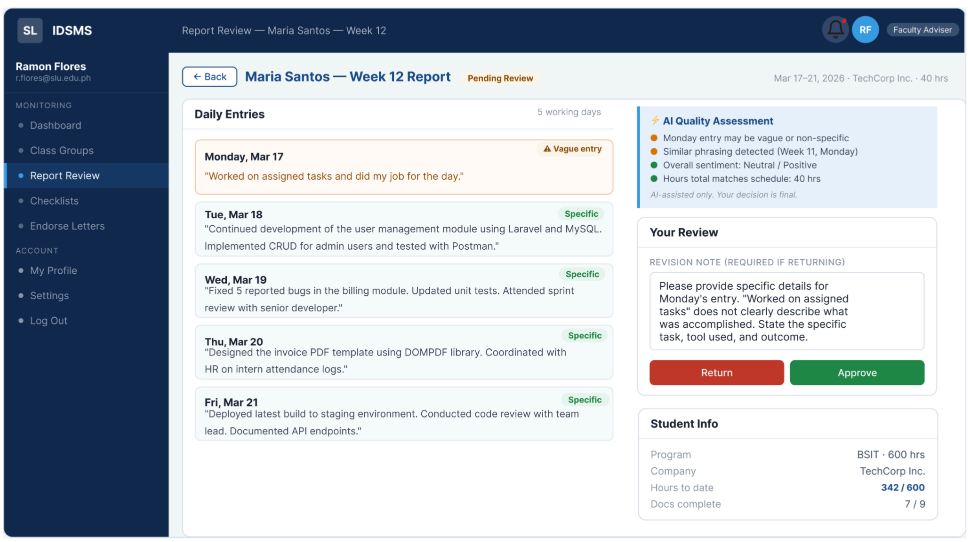

Figure 21: Faculty - Report Review page

Description	The Report Review page allows the faculty adviser to assess a student's weekly report. It shows, for each day, the accomplishment entries marked as Specific or Vague. The AI Quality Assessment panel indicates possible weaknesses (vague entries, reused phrases from previous weeks, sentiment, and hours accuracy). The faculty adviser records a revision note if they decide to return the report and then presses either 'Return' or 'Approve.' Student data (program, company, hours to date, and docs complete) is presented for the faculty as a point of reference.
Pre-Condition	The student should have submitted the weekly report. The faculty adviser should log in, and the student must be a member of the class groups assigned to the faculty member. The

	report should carry the status 'Pending Review.'
Process Required	1. Faculty goes to 'Report Review' or selects 'Review' from the Dashboard. 2. The system displays the student's report with daily entries and AI evaluation. 3. The faculty examines the AI side and each daily entry. 4. To approve the report: Faculty press 'Approve.' The report status is set to 'Approved,' and the student is informed. 5. To return the report: Faculty draft a revision letter and press 'Return.' The report status is set to 'Returned,' and the student is informed to edit.
Document/s produced	None

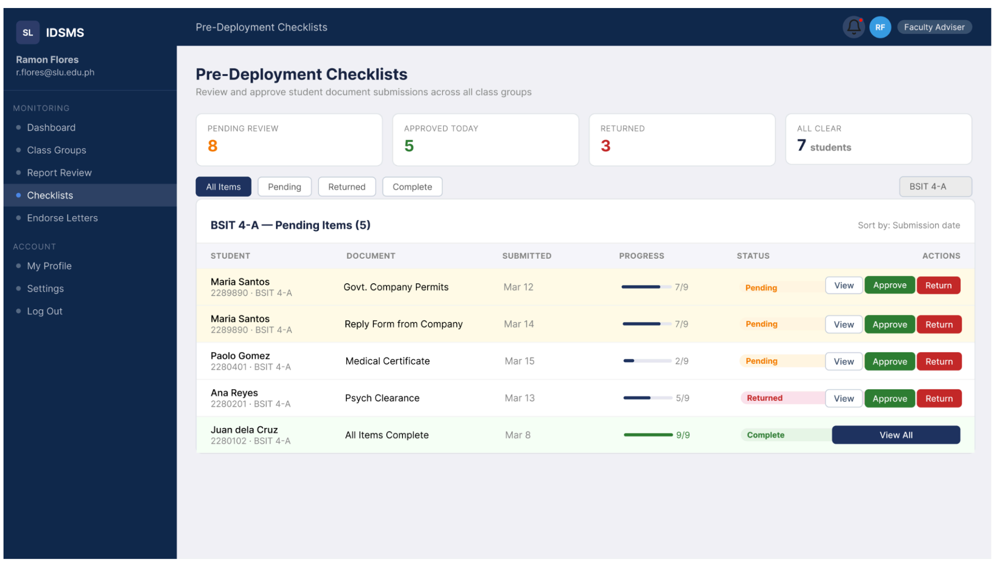

Figure 22: Faculty - Checklists page

Description	The Pre-Deployment Checklists page allows the faculty adviser to review and take action on documents submitted by students before deployment. A summary header shows the number of Pending Reviews, Approved Today, Returned, and All Clear students. The item list includes the student name, document name, submission date, progress bar, and status. Action buttons such as View, Approve, and Return are available for each item. Filter tabs include All Items, Pending, Returned, and Complete.
Pre-Condition	The faculty adviser needs to be logged in. Students must upload their pre-deployment documents with a 'Pending' status. These documents must be accessible in the system.

Process Required	1. The Faculty navigates to 'Checklists' from the sidebar. 2. The system displays all pending document submissions from assigned class groups. 3. The Faculty clicks 'View' to check the uploaded file. 4. The Faculty clicks 'Approve' to approve the document. The faculty clicks 'Return' to send it back to the student. 5. Status updates are shown immediately, and the student is notified.
Document/s produced	None

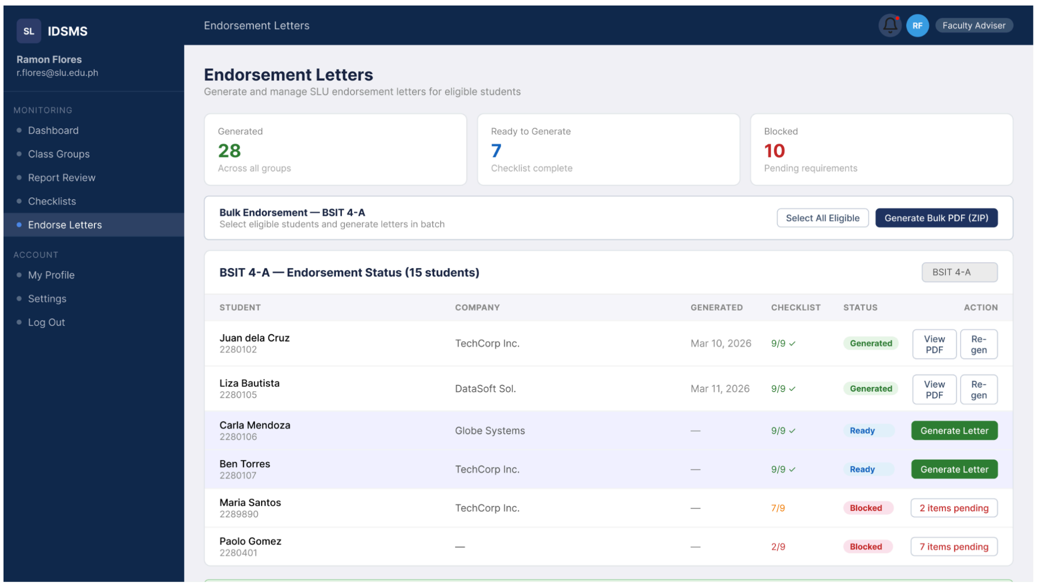

Figure 23: Faculty - Endorsement Letters page

Description	The Endorsement Letters feature located on this page is intended for the Faculty Advisor to create and manage SLU Internship Endorsement Letters for students fulfilling eligibility criteria. A summary lists the total number of letters generated, the number of students eligible for letter generation, and the number of students who have been blocked from generating letters for not fulfilling the eligibility requirements. Students who have completed their checklist will display a Ready status with a Generate Letter option. Students who have not completed their eligibility requirements will be displayed as blocked.
Pre-Condition	The Faculty Advisor must sign in, and the Student must have completed all of the pre-deployment Checklist items (9 of 9) to be eligible for the generation of an endorsement letter. The system must also contain an endorsement letter template configured.
Process Required	1. From the sidebar, navigate to the endorse

	letters page. 2. The system will check the completion status of each student's checklist. 3. For students who have completed their checklist and are eligible to receive an endorsement letter, the faculty will click the Generate Letter option to create the Endorsement Letter PDF. 4. To bulk produce the endorsement letter PDF for a class group, the faculty Advisor will select all students eligible to receive an endorsement letter and select the Generate Bulk PDF (ZIP) option. 5. All Endorsement Letter PDFs that have been generated will be stored in the system for download.
Document/s produced	Each eligible student will receive one Endorsement Letter (PDF), and the Bulk Export option will create a ZIP archive of the Endorsement Letters for all students in the entire class group.

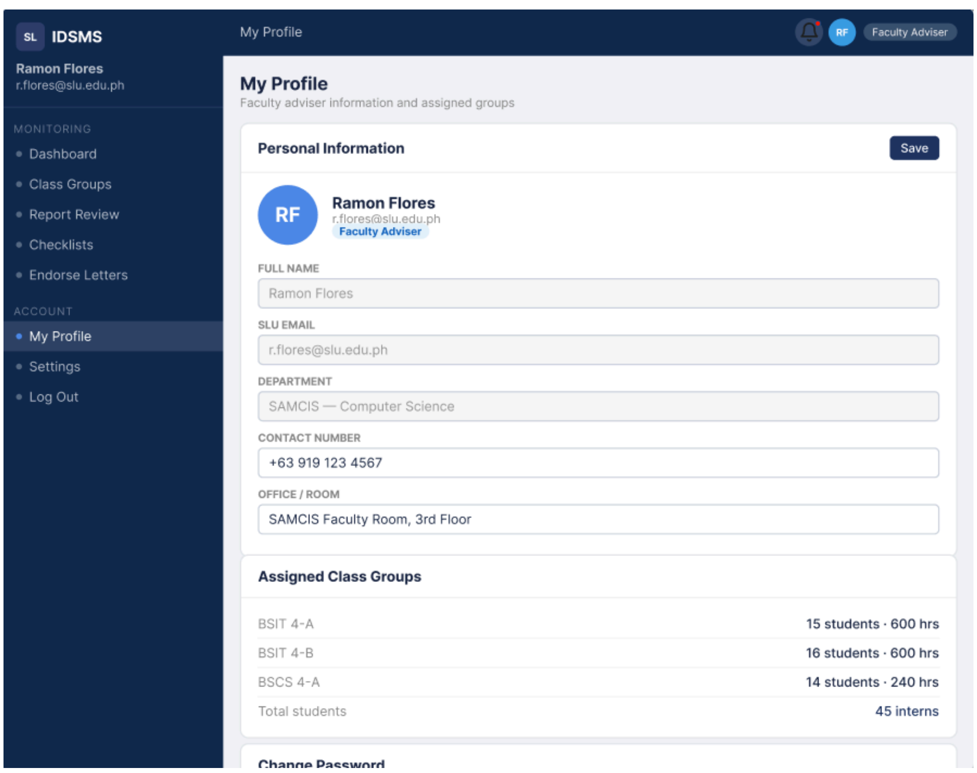

Figure 24: Faculty - My Profile page

Description	The Faculty My Profile page contains information such as full name, SLU e-mail, department, contact number, office/room, a list of assigned class groups with the number of students per group and the required hours for each group, and a Change Password section. The profile also includes a Save button that updates editable fields.

Pre-Condition	The faculty adviser must be logged in. The system administrator must have established the department and class group assignments.
Process Required	1. The faculty finds My Profile under the Account section. 2. The faculty can update the contact number and office/room. 3. The faculty would click the Save button to save the changes. 4. The faculty cannot change department information or class group assignments as they are read-only.
Document/s produced	None

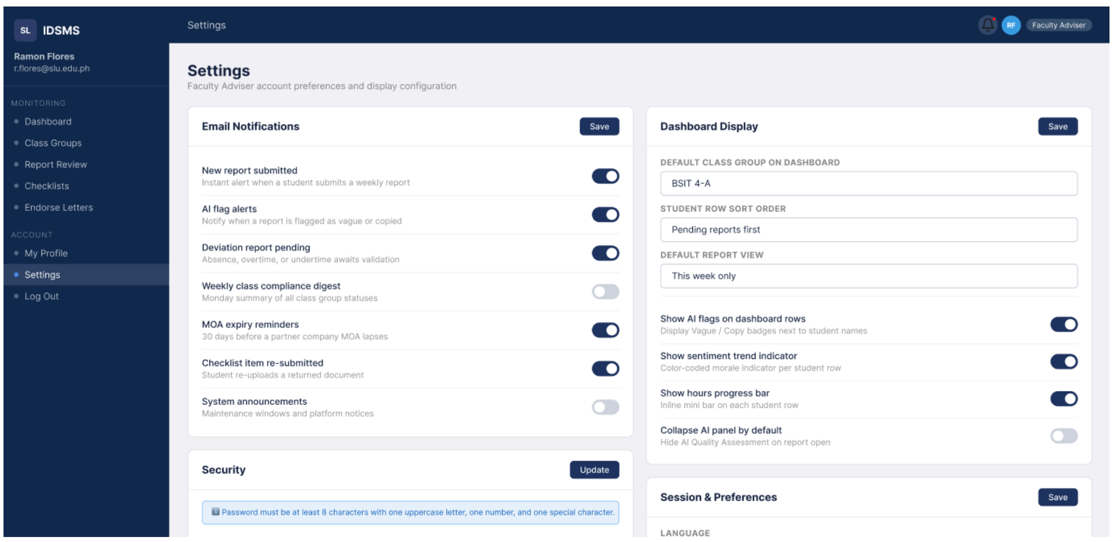

Figure 25: Faculty - Settings

Description	Faculty advisers can change the configuration of email notifications such as (newly submitted reports, AI flags, deviation reports that are waiting for approval, weekly compliance summary reports, reminders for MOA, re-submitting checklists, and system announcements) and the appearance of their dashboards (the default classroom group, student row sort order, default report view, whether or not to display AI flag, whether or not to display sentimental trend indicators, and how many hours of progress bars) from within the main navigation of the IDSMS. They can also manage their security settings and session preferences.
Pre-Condition	Faculty advisers must be logged into the IDSMS to accomplish these tasks.
Process Required	1. The faculty adviser goes to "Settings" in the left-hand side bar of the IDSMS. 2. The faculty adviser toggles all of their

	desired email notifications, storing them by selecting the "Save" button next to the section. 3. The faculty adviser configures all of the necessary dashboard display options, including the default classroom group, sort order of students, and whether or not to display sentimental trend indicators. 4. The faculty adviser can change his or her password in the Security section of the IDSMS if necessary.
Document/s produced	None

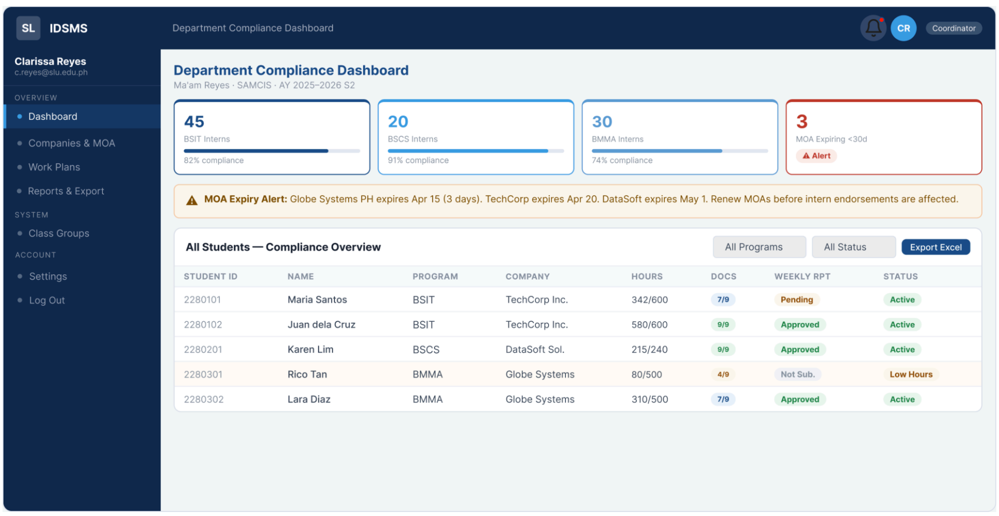

Figure 26: Department Compliance/Coordinator - Dashboard

Description	The Compliance Dashboard provides department-wide compliance information concerning the following programs: Bachelor of Science in Information Technology (BSIT), Bachelor of Science in Computer Science (BSCS), and Bachelor of Multimedia Arts (BMMA). The Compliance Dashboard is organized in two sections: several summary cards displaying the number of interns by program and compliance percentages, as well as an alert for the expiration of the Memorandum of Agreement (MOA) with partner companies; and a main table displaying all students, indicating Student ID, Name, Program, Company, Hours, Documentation, Weekly Report Status, and Status. Additionally, the Compliance Dashboard allows information to be exported to Excel for report creation.
Pre-Condition	The Compliance Coordinator must have a username and password with the Coordinator role in the system. In order to see students, the students must be currently enrolled for the active semester within the compliance system. The compliance system must be set up for

	both companies and MOA expiration dates.
Process Required	1. A Coordinator logs into the system and is automatically taken to the Department Compliance Dashboard 2. The system collects compliance data for each program and provides the coordinator with program-level summary information. 3. The Coordinator reviews the alert for any expiring MOAs and takes action to renew them. 4. The Coordinator filters the student table by program or status in order to find a specific student. 5. The Coordinator may export the entire compliance overview report to Excel for purposes of reporting.
Document/s produced	The Compliance Overview Report (Excel/CSV) will provide a summary of compliance information for all students the Compliance Coordinator sees across all programs.

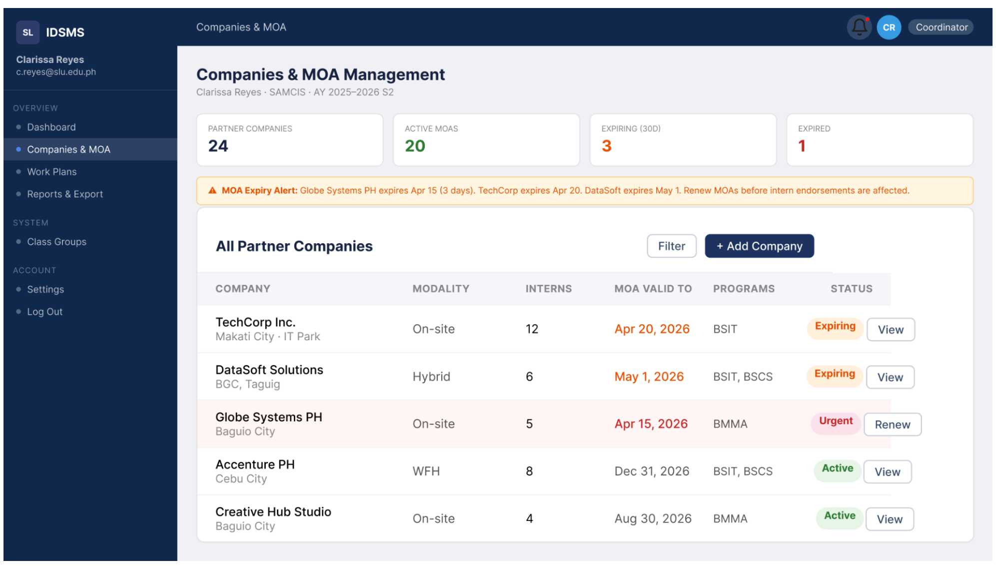

Figure 27: Department Compliance/Coordinator - Companies and MOA page

Description	The Companies and MOA Management section enables coordinators to effectively oversee and govern all partner companies and their associated Memorandums of Agreement (MOAs). This page contains Summary Cards that list the total number of partner companies, the total number of active MOAs, the total number of MOAs set to expire within 30 days, and the total number of expired MOAs. Additionally, a table will show all registered partner companies, along with their modality, intern count, MOA expiration date, covered programs, and current status (Active,

	Expiring, Urgent, or Expired). The actions available for each partner company are: View and Renew. There is also a '+ Add Company' button for this purpose.
Pre-Condition	Regardless of how often the coordinators use the Companies & MOA Management software, they must be logged in, have at least one partner company registered, and have an active semester configured in the system prior to using it.
Process Required	1. Using the sidebar, navigate to 'Companies & MOA'. 2. The system will display all partner registered companies and their respective MOA current statuses in real-time. 3. If an MOA is marked Expiring or Urgent, click 'Renew' to conduct the renewal process. 4. To view full details about individual companies (including the associated MOA documents), the coordinator will click 'View'. 5. To register a new partner company, click '+ Add Company' and fill in the company and MOA information, which will be saved in the system.
Document/s produced	The software can attach or link MOA files related to each company; however, no physical/bound version of any MOA or company information will be produced by the software's output. Instead, the software will maintain all the record information related to the registration of a partner company and the MOA electronically.

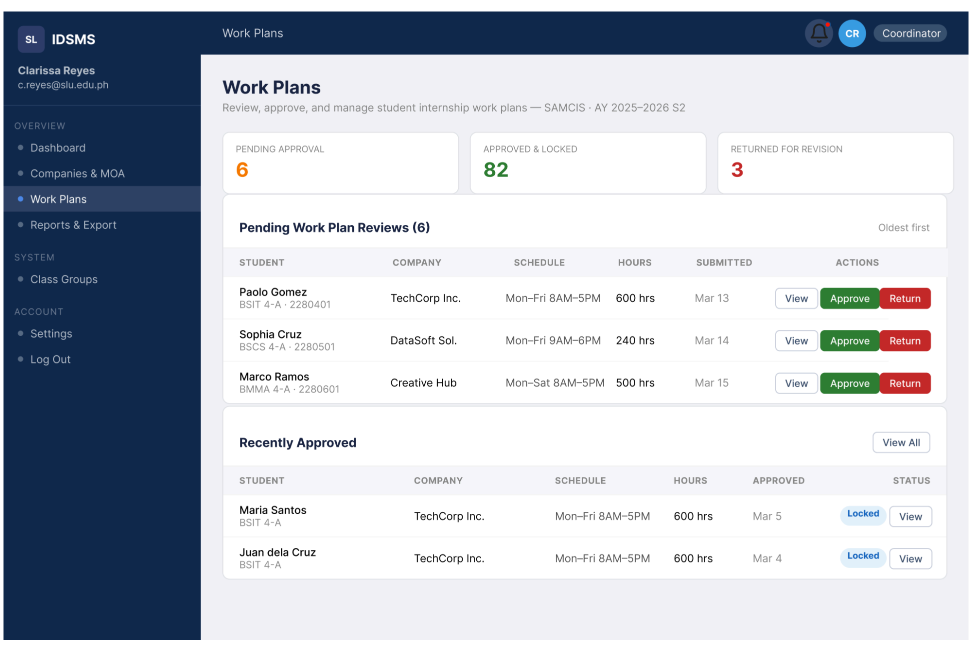

Figure 28: Department Compliance/Coordinator - Work Plans page

Description	The Work Plans page enables coordinators to verify and authorize student work plan submissions during internships. The header of this page provides information about Pending Approvals, Approved Plans (Locked), and Returned for Revision. Students' pending submissions are identified by name, program, company name, working schedule, required hours, submission date, and action buttons (view, approve, return). The most recent approved and locked work plan(s) are located on the right side of this page.
Pre-Condition	Coordinators must be logged into the system. Students must submit work plans in the system. Coordinators must have access to group/class programs for which they are responsible to review work plans.
Process Required	1. Coordinator accesses the 'Work Plans' option from the left-hand side of the sidebar. 2. The system will display all pending work plan submission(s) sorted by the submission date. 3. The coordinator will select the 'View' option to view the work plan submitted by the student. 4. The coordinator will select the 'Approve' option to approve/lock the work plan (will display 'Approved & Locked') or the 'Return' option to return the work plan to the student for revision. 5. The approved work plans will show in the locked status, and the students will receive notification.
Document/s produced	When a work plan is approved/locked, it becomes a locked record in the system. There will not be a separate document created for the approved work plan. Instead, the work plan form itself becomes the locked record when approved.

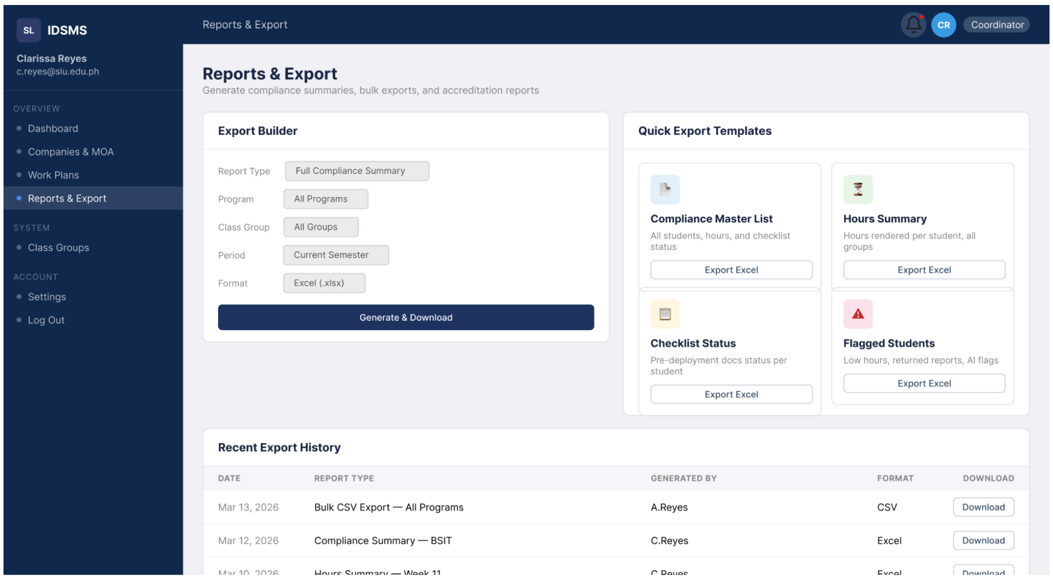

Figure 29: Department Compliance/Coordinator - Reports & Export page

Description	A compliance report can be generated and downloaded in various formats on the Reports & Export page. To generate a report, the Coordinator will use the Export Builder to select the report type (Full Compliance Summary), program, class group, period, and file type (Excel versus CSV) for the compliance report. Coordinators can also use the Quick Export Templates to export (1-click) the following reports: Compliance Master List, Hours Summary, Checklist Status by student, and Flagged Students by student. Each previously generated compliance report will be displayed in the Recent Export History table, with the ability to download any report.
Pre-Condition	The Coordinator must be logged into the system. Therefore, to generate a compliance report, the data for the selected semester and program must be available in the system (students enrolled, submitted hours, etc.). In addition, a report period must be active.
Process Required	1. The Coordinator will click on 'Reports & Export' in the sidebar 2. The Coordinator will configure the Export Builder parameters (report type, program, class group, period, file type) 3. The Coordinator will click 'Generate & Download' in order to generate and download the report 4. The Coordinator can also use a Quick Export Template by clicking on 'Export Excel' or 'Export CSV' associated with the desired report template 5. The generated compliance reports will be listed in the Recent Export History for future downloads of these reports.

Document/s produced	Full Compliance Summary, Hours Summary per Student, Checklist Status per Student, Flagged Students Listing in either Excel (.xlsx) or CSV format.

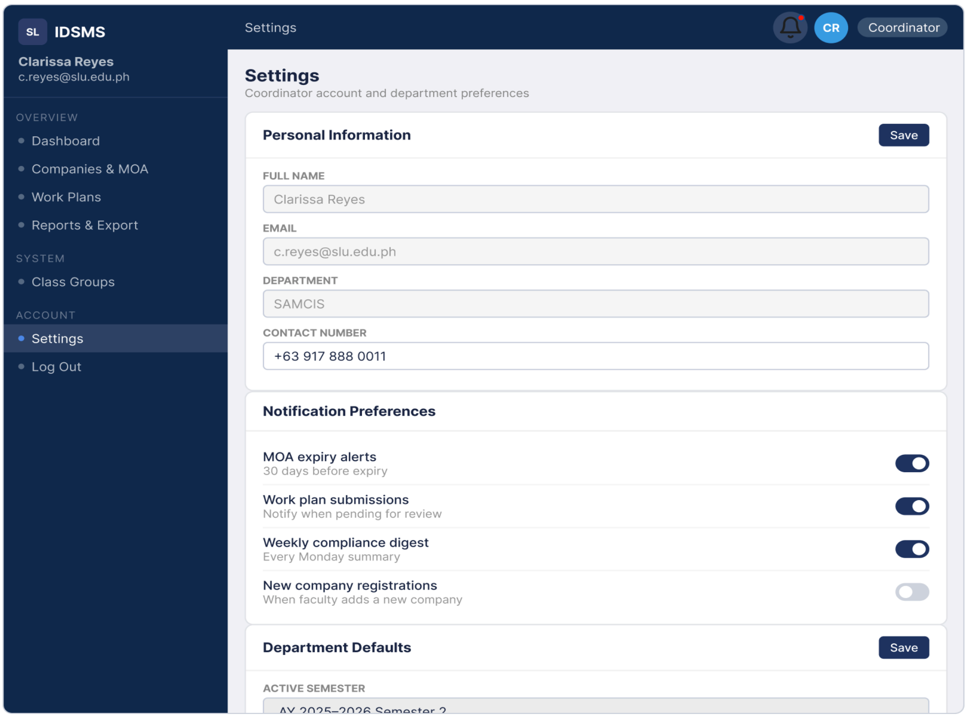

Figure 30: Department Compliance/Coordinator - Settings

Description	On the Coordinator Settings page, this is where the user will manage the following items: Full Name, Email, Department, Contact Number, Notification Preferences (i.e., MOA Expiration Alerts, Work Plan Submissions, Weekly Compliance Digest, New Company Registrations), and Department Defaults (i.e., Active Term, Any Other Defaults for All Coordinators' Departments).
Pre-Condition	The Coordinator must be logged in to their active Coordinator account.
Process Required	1. Coordinator selects the 'Settings' option from the left sidebar. 2. Coordinator enters updates to their personal information, then selects 'Save'. 3. The coordinator creates their Notification Preferences using the Toggle Switches. 4. Coordinator sets their Department Defaults (i.e., Active Term), and then selects 'Save'.
Document/s produced	None

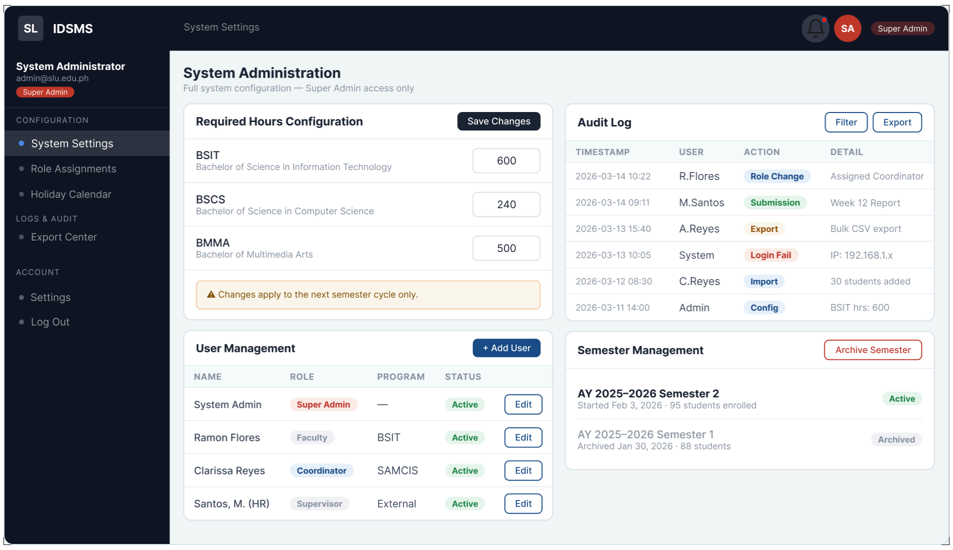

Figure 31: Super Admin - System Settings

Description	Access to the System Administration page is available only to the Super Admin role (the Super Admin is responsible for all system configurations). Each Required Hours Configuration per program (BSIT = 600 hours; BSCS = 240 hours; BMMA = 500 hours) is also listed, along with a User Management Table for managing users (adding users, editing user roles, viewing user status). An Audit Log is also available for all system events (which have timestamps), and there is a Semester Management area for archiving old semesters and activating new semesters.
Pre-Condition	Super Admin must be logged in as a Super Admin. The system must be powered ON, and the DB must be accessible.
Process Required	1. The Super Admin logs into the system and navigates to "System Settings." 2. The Super Admin sets the required number of hours for each program and clicks Save Changes 3. The Super Admin has full capability to manage users from the User Management Table within the system by adding new users, editing existing user roles, and activating/deactivating users. 4. The Super Admin can review the Audit Log for all system event records and export the data if necessary. 5. The Super Admin can archive all completed semesters and activate new semester cycles, using the Semester Management area.
Document/s produced	Audit Log is exportable in CSV or Excel format; no other documents will be produced; all Configuration Changes will automatically

	be reflected in the system's DB.

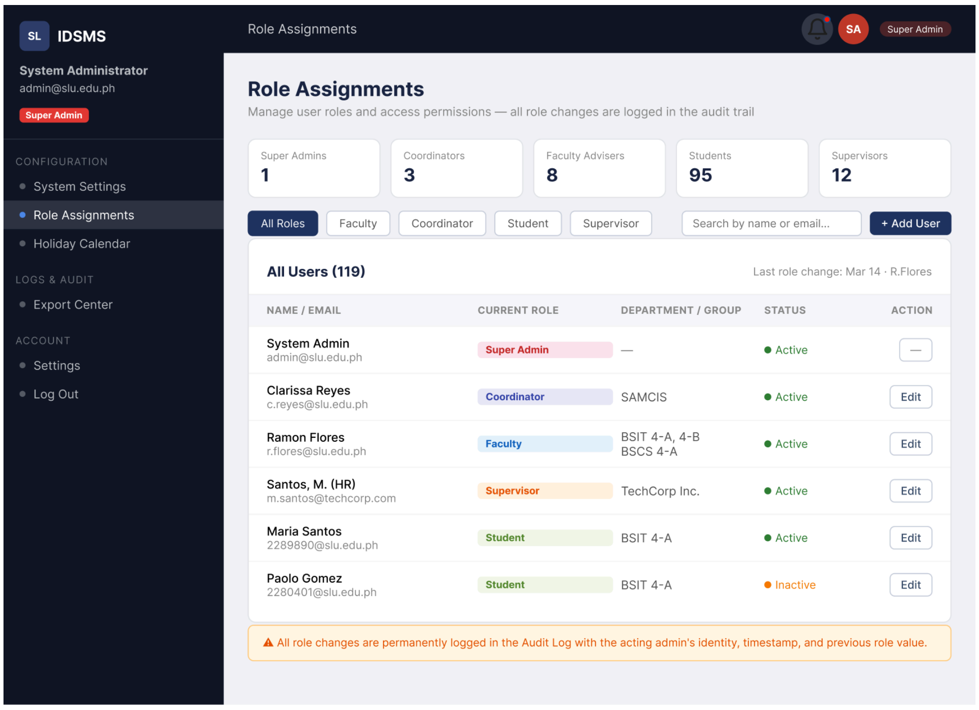

Figure 32: Super Admin - Role Assignments page

Description	Super Admins on the Role Assignments Page will have the capability to manage all IDSMS users and assign their roles (Super Admins, Coordinators, Faculty Advisers, and Students). The summary bar displays the total number of users for each role type. A searchable, filterable table lists all 119 users, and an Edit option is available. Any modifications to Users' roles and/or statuses are permanently recorded in the Audit Log. Hitting the "+ Add User" button allows you to add new users to the system.
Pre-Condition	Only Super Admins with their credentials can access this part of the system. Users should already be configured in the system, either existing in Active Directory or manually created or imported.
Process Required	1. The Super Admin chooses Role Assignments from the main menu. 2. The Super Admin uses filters or sorting to narrow down the selection of Users by their roles or search for the Users by name or email. 3. To make a change to a User's role, department/group assignment, or status, the Super Admin is required to click Edit next to the User. 4. The Audit Log will record any changes made to a User automatically, along with the

	Super Admin's identity and time stamp. 5. By clicking + Add User, the Super Admin will have the possibility to set up a new User account and give them a role.
Document/s produced	None

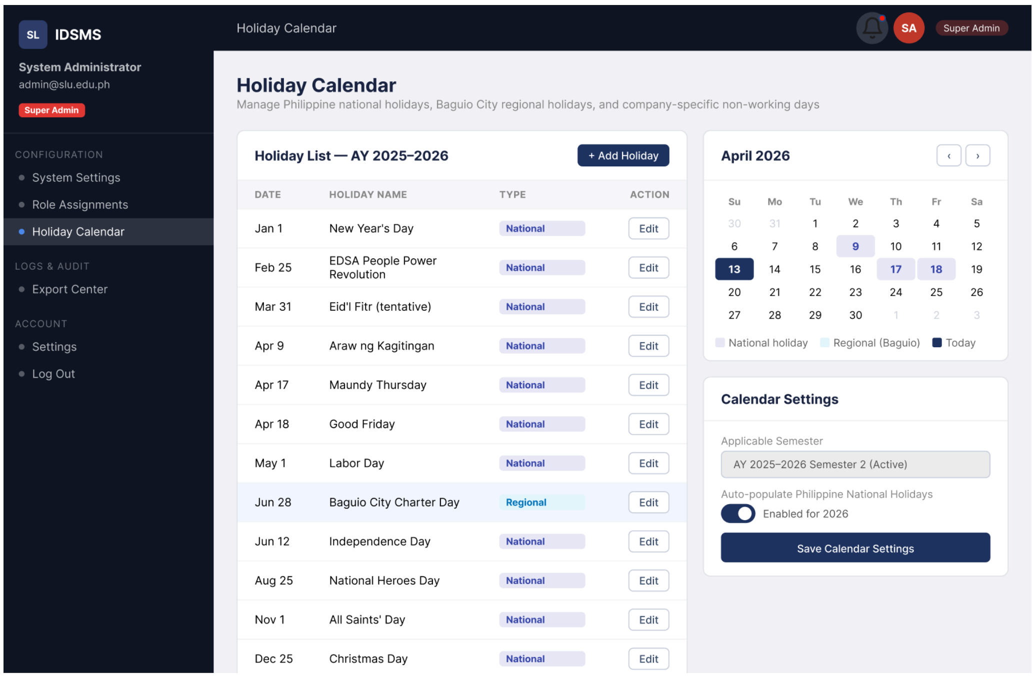

Figure 33: Super Admin - Holiday Calendar page

Description	The Holiday Calendar is a page that a Super Admin can use to organize Philippine national holidays, Baguio City regional holidays, and the company's own non-working days for the current semester. The holiday list displays each date, holiday title, whether it is a National or Regional holiday, and an Edit option. The calendar marks holidays and the current date. Calendar Settings allow the admin to select the semester and toggle auto-filling of Philippine national holidays.
Pre-Condition	The Super Admin should be logged in. The active semester should be set in System Settings.
Process Required	1. Super Admin goes to 'Holiday Calendar.' 2. Admin looks at the current list of holidays for the semester. 3. 3. Admin clicks '+ Add Holiday' to add new entries (date, name, type).4. 4. Admin changes existing holidays if necessary. 5. Admin switches on/off the auto-population of Philippine national holidays and presses 'Save Calendar

	Settings.' 6. Holiday information is updated in all the systems of attendance and hours tracking of the students.
Document/s produced	None

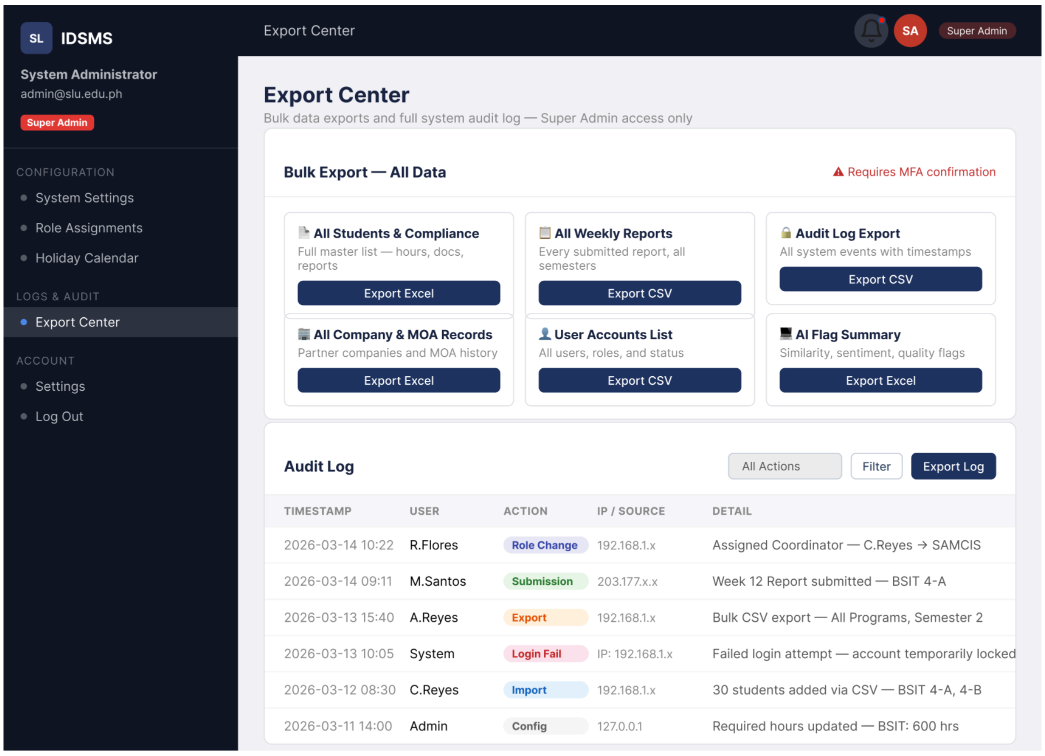

Figure 34: Super Admin - Export Center page

Description	Export Center is a page that allows Super Admins the facility to extract bulk data and access a comprehensive system audit log. Different bulk export formats are: All Students & Compliance (Excel), All Weekly Reports (CSV), Audit Log (CSV), All Company & MOA Records (Excel), User Accounts List (CSV), and AI Flag Summary (Excel). Multi-factor authentication (MFA) confirmation is necessary for all bulk exports. Audit Log displays records of all system activities, including time, user, action type, IP/source, and details, and is filterable and exportable.
Pre-Condition	The Super Admin must be logged in via MFA. The system should contain the data for the current semester.
Process Required	1. Super Admin opens 'Export Center.' 2. Admin chooses the bulk export type and presses 'Export.' 3. The system asks for MFA confirmation before continuing with such exports.

	4. Export file becomes available for download. 5. Admin can also select criteria for the Audit Log and then export the selected log via 'Export Log.'
Document/s produced	Bulk export types: All Students & Compliance (Excel), All Weekly Reports (CSV), Audit Log (CSV), Company & MOA Records (Excel), User Accounts List (CSV), AI Flag Summary (Excel).

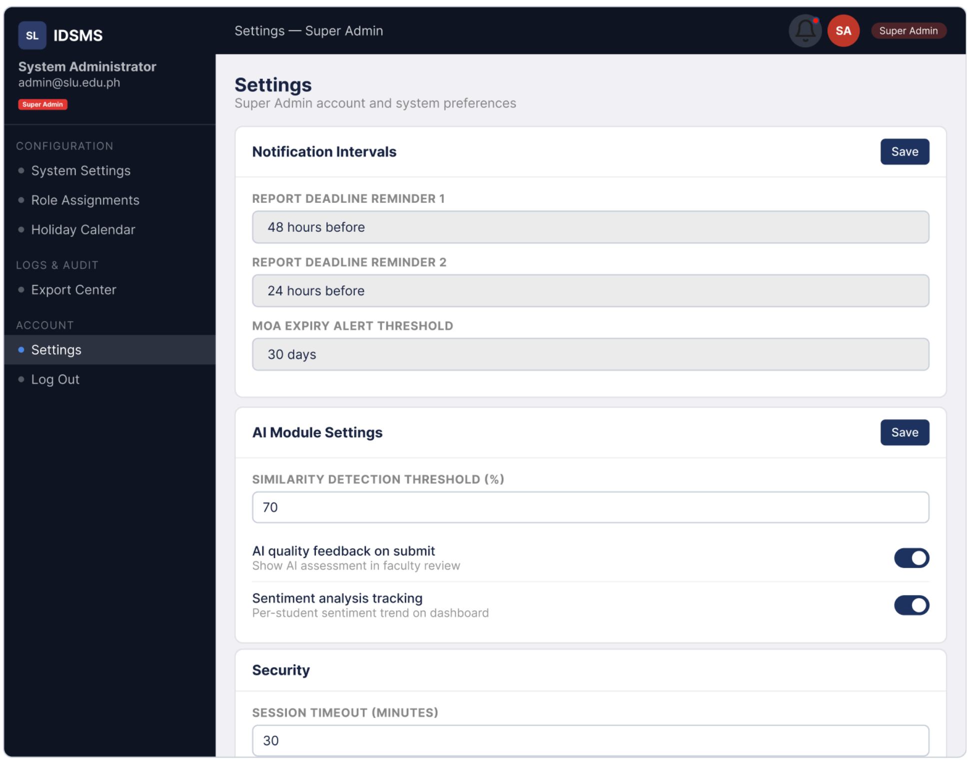

Figure 35: Super Admin - Settings

Description	The Super Admin Settings page sets up system-wide notification intervals (for example report deadline reminders sent at 48 hrs and 24 hrs prior to due, MOA expiry alert at 30-day threshold), AI Module Settings (e.g. similarity detection threshold at 70%, turning on/off AI quality feedback on submit, turning on/off sentiment analysis tracking), and Security settings (session timeout in minutes, at present, 30 minutes).
Pre-Condition	Super Admin must be logged in. The system should be up and running.
Process Required	1. Super Admin goes to 'Settings' from the Account menu. 2. Admin changes the values of the notification interval and presses 'Save.' 3. Admin sets the AI module parameters

	(similarity threshold, toggles) and presses 'Save.' 4. Admin chooses security options (session timeout) and saves.
Document/s produced	None

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APPENDICES

Appendix A: Interview Guide for Student Interns

Section	*Questions*
Student Submission	1. What documents or reports are you required to submit before and during your OJT? To whom do you submit them? 2. How do you submit your requirements (physically, email, Google Classroom, etc.)? 3. Is there a fixed schedule for submissions (weekly, monthly, end of OJT)? 4. What happens if you miss a submission deadline? Who do you notify and what is the next step? 5. Before submitting your documents to your faculty adviser, do you need to have them endorsed or signed by your company supervisor first?
Faculty Review	6. After submission, how long does it typically take for your faculty coordinator to review your report? 7. How does your faculty coordinator communicate feedback to you? 8. If your submission is incomplete or has errors, what does the review and revision process look like? 9. Does the faculty coordinator reach out to your company supervisor during review?
Approval Process	10. Who has the final authority to approve your OJT completion? 11. Is there a Memorandum of Agreement (MOA) or any formal document between SLU and your host company that needs to be signed before your deployment? 12. Does the company need to issue formal approval or certification before the school grants final approval? 13. What conditions must be met before approval is granted (hours rendered, reports, etc.)?

Report Generation	14. What types of reports are generated at the end of your OJT? 15. Who is responsible for generating each report (student, company, faculty)? 16. Where does the final report go after generation? 17. Is a summary or record of your OJT performance kept by the company for future reference?

Appendix B: Summary of Key Findings from Student Intern Interviews

Aspect	*Student A*	*Student B*	*Student C*	*Common Themes*
Submission Platform	Google Classroom for weekly/monthly reports	Google Classroom	Google Classroom	Heavy reliance on Google Classroom
Pre-deployment Documents	Company signatures, MOA-related	OJT waiver, medical cert, psych clearance	MOA processing (major bottleneck)	Multiple pre-deployment requirements
Weekly Report Process	Template from school, signed by company, submit weekly (before Monday)	Weekly every Tuesday	Weekly (Tuesday deadline)	Weekly submission with company signature
Late Submission Handling	Coordinate with faculty; possible extension + penalty	Private comment in Google Classroom	Screenshot proof + valid reason required	Informal workarounds; penalties common
Faculty Review Turnaround	1-2 weeks	2 days to 1 week	Within a week (private comments)	Inconsistent and delayed feedback
Hours Computation	Student computes; errors lead to penalties	Student computes	Student computes with proof	Manual computation prone to errors
Final Requirements	Summary report + company certificate	Technical report, certificates, evaluation	Technical report + 600 hours	Final report + proof of hours

Pain Points	MOA delays, late submission penalties, manual resubmission	Closed submission bins, private comments only	MOA processing, revision loops	Fragmented process, lack of visibility, manual workload

Appendix C: Key Interview Excerpts

C.1 Interview with Student A

Date: May 12, 2026

Interviewer: Frencine Daine Abanador

Key Excerpts:

The documents required before deployment include signatures, review of auditions, and other pre-deployment requirements from the company. After SLU gives the signal to start the OJT, we submit weekly reports using a template provided by the school. We fill in what we did during the week, have it signed by our company supervisor, and submit it via Google Classroom.

Google Classroom is mainly used for weekly and monthly reports. Confidential documents like medical certificates are still submitted outside the system.

Weekly reports are usually due before Monday. If a submission is missed, we coordinate with the faculty representative. Late submissions may result in penalties such as an extension of one week or additional requirements. I once had a miscalculation in rendered hours and had to extend my OJT by one week as punishment.

Faculty review usually takes one to two weeks. Feedback is given through private comments in Google Classroom. If there are errors or incomplete submissions, we must resubmit corrected versions, sometimes affecting previous reports. In one case, a single mistake required resubmitting multiple reports.

Final approval involves both the faculty coordinator and the company. The company issues a Certificate of Completion signed by the CEO. On the SLU side, we submit a summary/final report based on all weekly reports. The final technical report and completion documents are kept by the school.

C.2 Interview with Student B

Date: May 12, 2026

Interviewer: Frencine Daine Abanador

Key Excerpts:

Before starting the OJT, we submitted hard copies to the school including the OJT waiver, medical certificate, and psychological assessment. Company requirements included confidentiality agreement, company policy, and intern contract. During the OJT, we submit weekly accomplishment reports every Tuesday and monthly journals.

All submissions are done through Google Classroom. If a weekly report is late and the submission bin is closed, we attach the revised file through private comments.

Faculty review time varies from two days to one week. Feedback is given via private comments in Google Classroom. If there are errors, the faculty leaves a comment and requires resubmission by attaching the corrected file in the comment section.

Final approval is handled by the faculty coordinator. We are responsible for computing our own rendered hours. Required documents for completion include the technical report, proof of attendance, certificate of completion, company evaluation, and weekly reports.

The final report is submitted only to the school. The company does not keep a copy of our summary report.

C.3 Interview with Student C

Date: May 14, 2026

Interviewer: Frencine Daine Abanador

Key Excerpts:

The most challenging part of pre-deployment was processing the MOA, which delayed our start. We submit weekly reports every Tuesday using a school-provided template. The report must be complete, including attachments for leaves or absences.

If a submission is late, we provide proof (e.g., screenshots of communication with the supervisor) and request consideration from the faculty. Feedback is given through private comments in Google Classroom, usually within a week. Revisions are required if there are inconsistencies (e.g., mismatched hours or dates). We must get the corrected report signed again by the supervisor if needed.

The faculty conducts company visits to verify our progress and align it with weekly reports. Final approval requires completing the required 600 hours and submitting all documents including the technical report and company evaluation. Both the company supervisor and OJT coordinator are involved in the final clearance.